Linnéuniversitetet
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Home sweet home

A case study on persuasive technology to promote usage of an m-health application by elderly living at home

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Abstract

Much of the developed world is experiencing an aging population. This requires society to adapt to take care of a growing elderly population and improve their quality of life. Today, mobile systems are available that makes it possible to monitor and improve health (m-health). Even though these systems could be immensely helpful for the elderly population, this has not been the primary demographic for the current m-health systems.

This case study aimed at examining how persuasive technology (technology for changing behavior and/or attitude) can be used to promote usage of m-health applications by the elderly. For this purpose, a theoretical framework for supporting m-health systems is proposed. This framework consist of persuasive technology (for motivation and support for fulfillment of human needs), knowledge of elderly issues in interacting with mobile interfaces, smartphone usability heuristics and the *Plan-Do-Study-Act* (PDSA) cycle (to support goalsetting and incremental progress).

To determine how persuasive technology can be used to motivate elderly and find effective strategies for this purpose, the case study examined health behavior, motivations for healthy behavior, attitude to health, general goal setting behavior, needs, preferences, technological experience and self-efficacy, as well as usage of m-health systems using qualitative and participatory methods. Methods used included semi-structured interviews, future workshop, revolutionary rapid prototyping and usability evaluation.

The findings of the interviews and the future workshop suggested the primary motivation for healthy behavior was derived from social aspects. Thus, the most effective persuasive strategies for the elderly likely target their need for social belonging and socialization.

Based on the findings, an interactive prototype was developed. The prototype proposed an m-health application with self-monitoring that implemented an elderly community around healthy behavior, with opportunities to earn digital rewards and challenge other users to competition. The interactive prototype was then used in a usability evaluation to gauge its usability by the elderly and revised in higher fidelity according to their feedback.

Keywords: persuasive technology, m-health, mobile user interfaces for the elderly, elderly user experience, PDSA

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1 Introduction

This chapter presents the background to the research setting, its current challenges, as well as previous research to tackle these challenges. The context of this case study will be clarified, along with the research gap that inspires the research question.

1.1 Background

As time goes on and humans continue to develop, people are living longer and can enjoy better socioeconomic conditions. But, this has also led to an aging world. The elderly population is on a rapid increase in many countries across the world, with Japan on the forefront with a 30% proportion of elderly in 2012 (WHO 2015). WHO (2015) projects that many countries will have a similar proportion to 2012 Japan by the middle of the century.

This rapid increase in the elderly population has revealed weaknesses in the design of the healthcare system in accommodating the needs of the elderly. The conventional healthcare system is often structured to diagnose and treat time-limited health conditions in an episodic approach rather than long-term that may be more beneficial for the elderly. There is also a lack of integrated care which leads to an uncoordinated system that puts the responsibility on the elderly patient to manage their care across different health workers and conditions (WHO 2015).

WHO (2015) recommends adopting an older-person-centered and integrated care model that emphasizes the needs and goals of the individual. It will promote the patient to be an active participant in care planning and self-management of health. This provides long-term and proactive care that is integrated across different aspects of the care system. Sveriges Kommuner och Landsting (2018) supports the move to a more tailored approach to individuals in contact with the healthcare system and states that care should take into account personality, preferences, attitude to health and health behavior.

WHO (2015) states that ICT (Information and Communication Technology) will play a vital role in the integrated and older-person-centered care model to provide home-based and tailored care.

WHO (2015) stresses the importance of behavioral interventions for older people with unhealthy diets, sedentary lifestyles or certain risk groups (such as osteoporosis) to reduce health risks. Physical activity has been shown to have many benefits that contribute to an increased life expectancy and quality of life by increasing physical and mental capacities. Motivating and encouraging exercise in the elderly is crucial to preserving and strengthening their mobility. This includes reducing barriers to being mobile by improving the accessibility of their environment, promoting positive attitudes to exercise, and providing community events that encourage exercise and social involvement (WHO 2015).

1.2 Related research

The research areas for this case study revolves around three fields of research: persuasive technology, m-health and user experience (UX) geared towards an elderly audience. As shown in Figure 1 the research area for this case study is carried out where these areas overlap and incorporates elements from each of these fields.

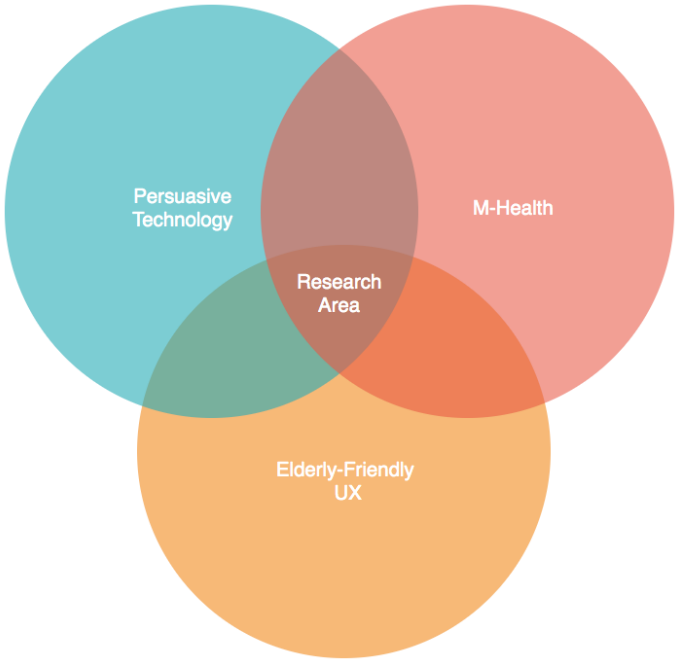


Figure 1 – Illustration of research areas

The literature search was performed using Google Scholar with the search terms: "persuasive technology" AND elderly AND mobile. Two relevant articles were identified. Through reading the title, abstract and keywords, they were deemed to coincide with the purpose of this case study and incorporated aspects from all three research areas. Below follows a brief overview of the selected articles.

Salim, Ali and Noah (2017) developed a persuasive mobile application for supporting a healthy diet for the elderly. The application enabled self-monitoring of Body Mass Index (BMI) and calorie intake. It also suggested menus. The application incorporated principles from the Persuasive Systems Design model by Oinas-Kukkonen and Harjumaa (2009). Principles incorporated included: reduction of complex behavior into simple tasks, process guides, information personalization, self-monitoring, interface attractiveness, high readability, clear language, similarity with the user, and trustworthiness (Salim, Ali & Noah 2017).

The application was evaluated via a 3-day usage trial ending with a questionnaire with three focuses: demographic information, user acceptance of technology, and persuasive features. This was followed up with semi-structured interviews with the

participants of the trial. Nine informants aged 50 to 59 and owned a smartphone were recruited via convenience sampling to take part in the evaluation of the application. The results showed that the application was received favorably by the informants and supported them well in monitoring their diet (Salim, Ali & Noah 2017).

In another study, Oliveira, Cherubini and Oliver (2010) investigated lacking medication compliance and declining compliance over time in the elderly. Oliveira et al. (2010, p. 251) hypothesized that "patients become more compliant in taking their medications when the task is not seen as an obligation, but rather as an entertaining and engaging experience".

To address the issue and prove the hypothesis, Oliveira, Cherubini and Oliver (2010) developed a mobile social game, MoviPill, supported by persuasive techniques. The MoviPill prototype registered date and time of when pill doses were taken by wireless sensors on the inside of the pillbox lid. This data is then validated server-side and updates the user's profile and status of the MoviPill social game. The players were awarded points depending on how accurate they were in the timing of dose intake as per the prescription. If the players were 15 minutes late in their dose intake time, MoviPill would trigger an alarm to remind them. MoviPill was tested during a six-week user study including 18 elderly participants. A control group was used were the participants had no reminders and were requested to press a button on the smartphone to register when they took a dose (Oliveira, Cherubini & Oliver. 2010).

Oliveira, Cherubini and Oliver (2010) states that the study shows that social competition can be an effective method to improve adherence to a prescription for independent elders. The MoviPill social game improved the dose intake behavior by higher timing accuracy of dose intake and remembrance for dose intake in comparison with the control group. The improvement was higher for elders that had an interest in games. Based on the findings, Oliveira, Cherubini and Oliver (2010) formulated nine design guidelines for addressing computing challenges for elders:

1. Use multimodal interaction (multiple modes/tools for interaction) for both input and output to account for different disabilities and needs.
2. Use walk-through and think aloud methods with the target audience to highlight low-visibility alternative task flows
3. Consistency is more important than bigger fonts and help text
4. Elders engage in competition thus game cheating should be prevented
5. The "right" competitors lead to higher motivation to compete
6. Rewards and persuasive strategies should be adapted to suit the specific user
7. Involve caregivers and family in the study as supportive observers
8. Elders are more open about sharing personal information when it is reciprocal
9. Consider the ecology where the technology will be used

1.3 Case study

The subject of this case study is the *From Empowering to Viable Living* (FRONT-VL) project. The aim of the FRONT-VL project is to develop an Internet of Things (IoT) system with the intent of helping elderly people to self-sufficiently live longer in their own homes and become more involved in their own health. The FRONT-VL project arose due to the new reality of a decline in the proportion of working-age people and longer life expectancy in Europe which warrants further support for self-sufficient living to reduce costs for healthcare and improve quality of life for the elderly (CELTIC-NEXT 2016).

FRONT-VL focuses on five use cases: *rehabilitation, fall prevention, mental health, palliative care* and *nutrition care*. The FRONT-VL IoT system target situations where mental, social and physiological impairments reinforce one another and form a negative cycle. The system intends to break this cycle and guide persons in these situations to a positive empowering cycle. To accomplish this, FRONT-VL aim to implement an open-source service framework of interconnected health-related ICT tools, IoT sensors and other platforms where the specific solutions can be adapted to the needs of the individual and results of big data analysis are available to the end-user (CELTIC-NEXT 2016).

In the context of the greater FRONT-VL project, this case study will focus on the mobile platform being developed by Sigma Technology that aims to visualize the data collected from IoT-sensors as well as built-in mobile sensors. The FRONT-VL mobile application is planned to provide opportunities for its users to set health goals and improve their health over time via continuous self-tests and self-monitoring as to make its users more involved in bettering their own health, support preventative care, decrease the occurrence of accidents and increase security of the user in their home.

The direct requirements for this case study, as given by Sigma Technology, is to develop a prototype of a mobile application graphical user interface that implements visualizations of sensor and self-test data, as well as goal setting and improvement planning via the plan-do-study-act (PDSA) cycle. The prototype is to act as a suggestion on how to design the interface and functionality of the FRONT-VL mobile application. Requirements include a user-centered approach and the inclusion of the target audience (the elderly) in the development of the prototype as to fulfill their expressed needs, goals and preferences. See Appendix A for the full summary of requirements from Sigma Technology.

1.4 Research gap

Orji and Moffat (2018) performed an empirical review of the last 16 years of literature on persuasive technology for health and wellness. Orji and Moffat (2018) concluded that there was a lack of involvement of the target audience in the design of the technology and therefore recommended a participatory design approach for future

research. The literature review also showed the need for persuasive technology to target a more diverse demographic such as older adults (Orji & Moffat 2018).

Seifert, Schlomann, Rietz and Schelling (2017) conducted telephone surveys to investigate the use of mobile devices for tracking physical activity by older adults (age 50 or older). The study showed that 20.5% of the participants in the surveys used mobile devices for tracking physical activity, with a higher likelihood if they were frequent exercisers, men, younger, or had an interest in technology. Main usage reasons were concluded to be self-motivational to adhere to healthy behavior rather than social aspects. Seifert et al. (2017) says that the results of the study and the potential benefits of this technology warrants further examination of elders use of mobile physical activity tracking to understand their motivations of use, as well as potential usability issues.

Together, these studies reveal that there is a gap in understanding of the motivations and intentions of the elderly population in the use of persuasive m-health applications. The study by Orji and Moffat (2018) suggests this may be because of a lack of proper involvement of this target audience in the design process of this type of technology. This may also have contributed to the need for further examination of usability aspects for older adults as expressed by Seifert et al. (2017).

1.5 Purpose and research question

The aim of the case study is to further the research on persuasive technology in the context of m-health applications for the elderly. A deeper understanding of the needs, preference, personality and goals of the elderly in relation to m-health and personal health may lead to advancements in persuasive strategies to change the attitude and behavior of the elderly individuals in this context. Integration of persuasive features and strategies targeting the elderly in m-health applications may aid the motivation for long-term use of the primary features of the application. This may also support the goal of FRONT-VL to break negative cycles that impact health and well-being, and progress to a more positive and empowering cycle. This might support elderly to self-sufficiently live longer at home with a lower incidence of accidents and medical issues. Based on this, the research question that is intended to be answered within this case study is:

How can persuasive technology promote usage of an m-health application by the elderly?

To answer this question, the case study intends to investigate how persuasive technology can support m-health applications, and find effective persuasive strategies targeting the elderly in order to promote continuous long-term usage of an m-health application which may enable self-sufficient health improvement.

1.6 Scope and limitations

This case study specifically aim to investigate: health behavior, motivations for healthy behavior, attitude to health, general goal setting behavior, needs, preferences, technological experience and self-efficacy, as well as usage of m-health systems of elderly people. This is done to gather data for determining effective persuasive strategies targeting elderly individuals and designing an m-health system with high usability and ease-of-use.

In addition, the case study intend to formulate a framework for persuasive m-health systems that may enable an overall positive user experience for the elderly in interacting with this kind of system.

A participatory approach will be used in the selection of data collection methods as to involve the users (the elderly population) in the design process of the m-health prototype. Data gathered through methods employed will directly influence the m-health prototype and its implemented solutions.

Implementation of specific self-test methods, analysis of specific health conditions or medical integrations (such as prescription management) in the m-health prototype won't be examined as it is not specifically relevant to the purpose of this case study.

This study won't directly investigate or take into account the impact of culture on the usage of m-health applications, attitude to persuasive technology, or the general view on health influenced by culture as this is outside the scope of this case study. The specific healthcare system used in the country of informants may as well impact the behavior and attitude of people, but this aspect won't be taken into account in this case study.

1.7 Target group

The target group for this thesis is primarily developers of m-health applications as an example of how persuasive technology can be used to provide further support of the main functions of the application to increase user retention and motivate continuous use. The thesis also targets researchers of persuasive technology as a contribution to knowledge on the topic of how to approach the elderly with persuasive strategies in the context of m-health.

The thesis may also be of interest to elderly people that use or has an interest in m-health applications, to gain understanding of the theories that motivate the inclusion of persuasive features.

1.8 Thesis structure

The thesis starts with the *literature review* chapter that gives a detailed explanation of the research areas of the case study: m-health, persuasive technology, and elderly UX.

It also details other literature and theories related to the case study. This includes the PDSA cycle. This chapter concludes in a synthesis of the theoretical framework which explains how the theories relate to each other as well as their benefit and conduct of use for this case study.

The following chapter is *methodology*, which explains the scientific approach underlying this case study, as well as chosen methodology for data collection, analysis and method execution. The empirical findings and analysis of these are thereafter presented in the *results* chapter. In the following chapter, *discussion*, the implications of the results are discussed and how it answers the research question and fills the research gap. The final chapter, *conclusion*, briefly summarize the case study and its findings as well as presents possible avenues for further research.

2 Literature review

This chapter presents the relevant literature that forms the theoretical framework for the case study pertaining to m-health, persuasive technology, mobile user interfaces and user experience toward an elderly target audience, as well as other literature relevant for the case study (PDSA).

2.1 M-health

M-health has been defined in at least three ways: “medical and public health practice supported by mobile devices, such as mobile phones, patient monitoring devices, personal digital assistants (PDAs), and other wireless devices”, ” mobile computing, medical sensor and communication technologies for health-care”, as well as ”emerging mobile communications and network technologies for healthcare systems” (Bravo, Hervás, Fontecha & González 2018).

Bravo et al. (2018) says m-health is a vital tool to support the well-being of the aging population through monitoring of physical, mental and social aspects and provide feedback to promote health improvement and personal health management. M-health applications targeting specific health aspects are bountiful, but Bravo et al. (2018) criticize the current approach of these applications of being too specific and requiring a high degree of interaction and cognitive effort to use. Bravo et al. (2018) says that human-computer interaction for these type of applications should preferably be approached in a “disappeared” way where the functional processes run in the background and require minimal user interaction to operate.

Bravo et al. (2018) summarize and analyze the experiences from m-health systems developed in case studies by the Modelling Ambient Intelligence (MAml) Research Lab to extract the lessons learned from these case studies.

Bravo et al. (2018) states that one of the main challenges of m-health is to support people with chronic diseases in their everyday life. Bravo et al. (2018) states that one solution for this is personalized health monitoring mobile applications. One such system was investigated in a MAml Research Lab case study, where personalized applications were generated using an ontological data model based on patient’s profiles and module definitions for the mobile phone. Monitoring data were then stored on a server for review by a doctor (Bravo et al. 2018).

M-health has been shown to be able to support obesity treatment via monitoring of physical activity, weight recording through Wi-Fi scales and diet recommendations based on gathered data and user characteristics. M-health has also shown promise for supporting gait analysis and frailty assessment as concluded by tests carried out by MAml Research Lab (Bravo et al. 2018).

M-health systems could be used to support elders in managing their medicine prescriptions. An m-health system developed by MAml Research Lab used control tags and NFC tags on medicine boxes that enabled the elder to interact with it to send a request for refilling the prescription to their family doctor. The system also enabled personalized prescription information through the NFC tags and the ability to send alerts (for example when treatment ends or when it is time to take the medicine). This system outmodes the traditional approach to prescriptions that require a doctor's appointment (Bravo et al. 2018).

Case studies by MAml Research Lab in the context of nursing showed that NFC-enabled smartphones can be used to access patient information through NFC tags placed on patients or rooms. The system also make it possible to transfer patient information from nurse to nurse through their smartphones. This system replaces the task of manually filling out forms with NFC-aided information transfer to more efficiently handle patient information so that nurses can focus on patients rather than data. Similarly, NFC-enabled smartphones and NFC tags and bracelets could be used to simulate nursing routines for educational purposes (Bravo et al. 2018).

Bravo et al. (2018) says that m-health is still in the early stages, but the interest in developing m-health systems for healthcare are increasing. As shown by the previous case studies, there are many application areas for m-health in supporting the healthcare system and well-being of individuals. This includes being a complement to existing care systems, monitoring of physical exercise, monitoring of vital signs, and remote healthcare (Bravo et al. 2018).

Bravo et al. (2018) propose a new approach to healthcare where m-health acts as a complement to traditional hospital healthcare services. With m-health, healthcare services can be provided even outside of healthcare facilities through remote healthcare and monitoring. With less patients in healthcare facilities, this may make it possible for more attention given to individuals. Though it is important that m-health systems are developed to include the entire health community and not merely the end-user, to combat lack of communication and isolation of patients (Bravo et al. 2018).

In the "disappeared" approach to m-health applications suggested by Bravo et al. (2018) one should strive for short, easy and non-distracting interactions where the monitoring aspects can autonomously run in the background during activities. Continuous monitoring is useful for prevention of health issues and analysis of user data provides possibilities for care personalization (Bravo et al. 2018).

Bravo et al. (2018) states that finding ways to develop easy interactions to promote a balanced diet for m-health is an important factor for supporting healthy lifestyles. With Cloud Computing, data for healthcare services can be securely stored and accessed anywhere. This as well enables opportunities for synchronization and collaboration. Cloud Computing also makes it possible to perform Big Data Analytics

to perform large-scale analytics to find patterns, form the base for a diagnosis or prediction and prevention of diseases. In handling patient data care must be taken to preserve privacy as well as security of the data storage (Bravo et al. 2018).

Bravo et al. (2018) points out that that gamification techniques can be used to encourage patient use of m-health. Integration of social networks may also enable information sharing and advice between peers which can encourage its use.

This approach by Bravo et al. (2018), as he points out, requires a large-scale change in health behavior and embracing of smartphones and m-health systems as a valuable component of the healthcare system. Currently, the use of m-health is undervalued in the healthcare system. To combat this view, it requires a focus on ease-of-use for both patients and caregivers (Bravo et al. 2018).

2.2 Persuasive technology

According to Oinas-Kukkonen and Harjumaa (2009) usage of persuasive technology for healthcare applications can be especially beneficial to motivate people toward healthy behavior and by result delay or prevent medical problems, which by extension can economically unburden the healthcare system. Though, Oinas-Kukkonen and Harjumaa (2009) criticized existing frameworks for being too limited to be used as a model for the development and evaluation of persuasive technology and thus developed the Persuasive Systems Design (PSD) model based on their empirical work, research and conceptual analysis in the field of persuasive technology. Oinas-Kukkonen and Harjumaa (2009) define persuasive systems as "computerized software or information systems designed to reinforce, change or shape attitudes or behaviors or both without using coercion or deception". To *reinforce* a person's attitude and/or behavior means to make them more resistant to changing their attitude or behavior. To *change* means to change their current attitude and/or behavior which changes their response to issues. To *shape* means to create new attitude and/or behavior patterns for a situation where none previously existed.

In the PSD model the development process consists of three steps described as *understanding key issues behind persuasive systems*, *analyzing the persuasion context*, and *design of system qualities*.

Oinas-Kukkonen and Harjumaa (2009) says that it is critical to reach an understanding of the issues inherent in persuasive systems before one can analyze and design such a system. Oinas-Kukkonen and Harjumaa (2009) has defined seven postulates that need to be understood in the first step.

The first postulate, *Information technology is never neutral*, describes that information technology is a continuous persuasive presence that affects people's attitude and behavior in specific directions. In addition, persuasion is a complex

process that must be able to adapt to that the user's goals and motivations can change during the process (Oinas-Kukkonen & Harjumaa 2009).

The second postulate, *People like their views about the world to be organized and consistent,* is based on the concept of *cognitive consistency*; if a person discover an inconsistency between their committed attitude and behavior, then *cognitive dissonance* may arise that motivates them to realign their attitude and behavior to restore consistency. This may be exploited by persuasive systems by enabling and encouraging users to make commitments to a target behavior as well as presenting information inconsistent with the user's way of thinking (Oinas-Kukkonen & Harjumaa 2009).

The third postulate, *Direct and indirect routes are key persuasion strategies,* clarifies the appropriate *route* for a *persuasive message* depending on the characteristics of the recipient. There are two different routes: *indirect* and *direct*. The direct route refers to persuasive messages that appeal to the individual's reason and intelligence. The indirect route uses simplistic cues to trigger heuristics in the recipient. Individuals that have the ability and motivation to evaluate the contents of persuasive messages are more recipient to the direct route and if motivation and ability is low the indirect route is more appropriate. The direct route has been shown to be more effective at persuasion. Though, usage of the indirect route may be unavoidable due to time-constraints or lack of ability and/or motivation to evaluate the message in certain situations (Oinas-Kukkonen & Harjumaa 2009).

The fourth postulate, *Persuasion is often incremental,* asserts that it is easier to reach a goal if it is broken up into an incremental process with a series of subgoals. Therefore, a persuasive system should enable the user to make incremental progress toward a target behavior but make the end-goal apparent throughout the process.

The fifth postulate, *Persuasion through persuasive systems should always be open,* stresses the importance of the persuasive system to be transparent. The bias of the designers should be available to scrutiny by the users and information provided by the system should be truthful. Neglecting to follow this may result in misleading the users or the system losing persuasiveness due to distrust (Oinas-Kukkonen & Harjumaa 2009).

The sixth postulate, *Persuasive systems should aim at unobtrusiveness,* explain that persuasive features shouldn't disturb users when performing tasks using the system or be triggered at inopportune times.

The seventh postulate, *Persuasive systems should aim at being both useful and easy to use,* emphasizes the importance of providing a good user experience, usability and ease-of-use in order for persuasive systems to be successful. If any of these aspects

are lacking, it may negatively affect the persuasiveness of the system (Oinas-Kukkonen & Harjumaa 2009).

In the second step of the PSD model the *persuasion context* is analyzed by reaching an understanding of the persuader, persuadee, message, channel and larger context of the persuasive system. This is critical to understand in order to effectively persuade the user. This step is broken down into three parts that analyze separate areas of concern: *The Intent*, *The Event* and *The Strategy* (Oinas-Kukkonen & Harjumaa 2009).

Oinas-Kukkonen and Harjumaa (2009) says *The Intent* is the first area that needs to be explored. This includes identifying the *persuader* and what *change type* that is targeted. The persuader is the source of the intention to persuade. There are three recognized sources of intentions: those who create the interactive technology (endogenous), those who distribute the interactive technology (exogenous), and the user of the interactive technology (autogenous). In endogenous technologies the importance of transparency is further amplified, and the technology has to be designed with the users' own volition to change their attitude or behavior in mind. In exogenous technologies it is important to include the possibility for personalization of assigned goals. Autogenous technologies should focus on creating a rewarding user experience to motivate long-term use for people who want to change their own attitude or behavior (Oinas-Kukkonen & Harjumaa 2009).

Change type refers to if the persuasive system aims to change attitude, behavior or both. Oinas-Kukkonen and Harjumaa (2009) says the theory of cognitive consistency suggest that it is easier to affect attitude through behavior change than affecting behavior through attitude change. If behavior changes, attitude change naturally follows. Attitude does not always correlate with behavior and it is possible to change a person's behavior with a persuasive system even though a positive attitude to the behavior is absent (Oinas-Kukkonen & Harjumaa 2009).

In *The Event*, one should consider the *use context*, *user context* as well as the *technology context* for the persuasive system. For the use context the problem domain should be explored as to determine required features. When it comes to persuasive systems for health, people often already have the correct attitude as well as the necessary information to act in line with their attitude, but bad habits reinforced over a long period have made it difficult to permanently change their behavior. Persuasive systems should in this example context make it easier for the user to adhere to the proper attitude even in difficult situations (Oinas-Kukkonen & Harjumaa 2009).

Oinas-Kukkonen and Harjumaa (2009) states that along with the use context, the user context needs to be analyzed. This includes a wide array of individual factors such as the user's needs, goals, motivations, abilities, interests, attitudes, culture, personality and more. The individual's need for challenging cognitive endeavors with the system

affects which persuasion route is effective. People with high need for cognition should be approached via the direct route, and the indirect route is more effective for people with a low need for cognition. The most critical factors to analyze in the user context is the user's goals, past attempts at reaching their goal and their current progress towards their goal (Oinas-Kukkonen & Harjumaa 2009).

Oinas-Kukkonen and Harjumaa (2009) states that the technology context is important and that one should determine the appropriate technological platform for the persuasive system. To do this, the strengths and weaknesses of each potential platform need to be explored.

Analysis of *The Strategy* requires a thorough investigation of the persuasion *message* and *route*. Oinas-Kukkonen and Harjumaa (2009) states that a persuasion situation can be defined as "an event in which the persuadee makes optimal compromises among conflicting forces". Persuasion can also be defined as changing attitudes and/or behavior. Persuasion often attempts to convince the persuadee, but there is a fine line between persuasion and convincing. Persuasion attempts to provoke emotions through symbolic strategies, while convincing targets the persuadee reason and intelligence with logical arguments (Oinas-Kukkonen & Harjumaa 2009).

The final step of the PSD model is design of system features. Oinas-Kukkonen and Harjumaa (2009) states that requirements specification is a cornerstone of software development and describes how the system should behave, qualities it must possess, as well as design and development constraints. The persuasiveness of a given system is mainly based on the qualities it possesses. The seven postulates highlight some aspects that need to be taken into account when designing a persuasive system, but further requirements need to be clearly defined to facilitate communication with stakeholders and developers of the system. In short, the steps to take in realizing a persuasive system includes: *analysis of persuasion context and selection of persuasive design principles*, *requirement definition for software qualities*, and *software implementation*. Oinas-Kukkonen and Harjumaa (2009) suggests four different categories of design principles for persuasive systems: *primary task*, *dialogue*, *system credibility*, and *social support*.

Primary task design principles support the primary task the user carries out in the system. Design principles for dialogue is based around system feedback and messaging that motivate and help users to work towards their goal. The design principles for system credibility explain how to design a system with higher credibility that by extension increases its persuasiveness. The design principles for social support explain how to motivate users by social influence (Oinas-Kukkonen & Harjumaa 2009).

2.3 Elderly-friendly user experience for mobile devices

According to ISO (2018), user experience (UX) is defined as "user's perceptions and responses that result from the use and/or anticipated use of a system, product or service".

This definition infer that constructing a positive UX is a matter of high complexity that isn't achieved by just one aspect of interaction, but all aspects in and around a system. As Hassenzahl (2008) says, experience is an ongoing event that is affected by continuous reflection of the experience; the experience is regulated by positive and negative emotions to the interaction that determines one's desire to continue or stop the interaction. According to Hassenzahl (2008), UX is a dynamic and temporal evaluation on a "good-bad" scale.

Hassenzahl (2008) says that to achieve good UX, interactive products need to facilitate the user's *do-goals* and *be-goals*. That is, the product need to have a *pragmatic quality* that supports the user's do-goals, such as "making a phone call". Therefore, the product is required to have a utility and usability for the user in performing tasks. It also need to have a *hedonic quality* that support be-goals, such as "being competent" or "being special". Be-goals focuses on the self and basic human needs thereof, for example need for self-expression, stimulation or personal growth.

Hassenzahl (2008) argues that it is the fulfillment of be-goals that is the driver of positive experiences and users will attach perceived hedonic qualities to products supporting this. Usability in itself is not a desired goal; it derives its value through facilitating the fulfillment of be-goals. As such, an interactive product need to support the fulfillment of do-goals, while keeping the underlying be-goals in mind.

2.3.1 Smartphone usability heuristics

Heuristic usability evaluation is a commonly used method to assure a products quality and its ability to properly enable users to efficiently reach their goals (Inostroza, Rusu, Roncagliolo, Rusu & Collazos 2016).

Inostroza et al. (2016) developed a set of 12 usability heuristics for smartphones called SMASH (SMArtphone's uSability Heuristics). SMASH was developed in a 6-step process consisting of: (1) collection of bibliography on smartphones and related usability heuristics, (2) highlighting of important parts of the collected information to establish the main concepts, (3) identify characteristics for smartphone heuristics, (4) specify the set of usability heuristics, (5) validate the heuristics against traditional heuristics and user tests, (6) refine based on feedback from validation step.

The heuristic SMASH1 specifies that users need to be informed of any state change in the system, whether they be through direct user action or triggered by the system. System feedback for smartphones for state change can include sound, animation,

messages, text notifications, light notifications, and vibration. Properly implemented this heuristic allows the user to react to events in an appropriate manner and gives them better awareness of the status of the system. When evaluating, one should also consider the difference between non-implemented “visibility of system status” and lack of feedback due to performance issues such as from low battery or too many running processes (Inostroza et al. 2016).

SMASH2 specifies that the system interactions should mimic real-world interactions as closely as possible. This means that system responses to user interactions should also approximate real-world responses. For example, swiping speed affecting scroll distance is an approximation of real-world inertia. Implementation of this heuristic lowers the barrier for entry for new users and minimizes errors due to utilization of recognizable concepts and language (Inostroza et al. 2016).

SMASH3 specifies that users should be able to easily rectify mistakes by aborting processes or exiting from other undesirable states to a desirable one. This gives the user better experience and efficiency of use with a higher sense of control over the system (Inostroza et al. 2016).

SMASH4 specifies that the system should follow set standards and conventions so that new users can rely on their prior experience in using similar systems. Consistent standards can range from different parts of the same system, devices from the same manufacturer, or devices using the same operating system (Inostroza et al. 2016).

SMASH5 specifies that options and functionalities need to be explicitly explained and further information should be available to the user according to their need. The user should be warned if an action is irreversible or might have undesired effect. For smartphones, where screen real estate is a concern, informational icons are important. Properly implemented, this heuristics can prevent and minimize errors and by extension provide a better efficiency of use (Inostroza et al. 2016).

SMASH6 specifies that one should strive to reduce the user’s memory load by making sure that relevant information is presented where it is actually used, and that complex instructions are simplified. This also reduces the mental effort required to use the system (Inostroza et al. 2016).

SMASH7 specifies that the system should provide customization options so that it can be personalized according to the user’s preferences and needs. This provides a better sense of ownership to the user, as well as efficiency (Inostroza et al. 2016).

SMASH8 specifies that the system need to load information in a reasonable time and minimize the steps required to perform tasks. Performance of software in relation to hardware should be considered to reduce load times. Animations should also be fluid and should not negatively impact the efficiency of use.

SMASH9 specifies that the design should strive for aesthetic minimalism and avoid displaying too much information at once. This leads to better performance due to fewer resources used as well as reduces stress and exhaustion from information overload (Inostroza et al. 2016).

SMASH10 specifies that the system should provide clear instructions on how to recover from errors to reduce the user's frustration (Inostroza et al. 2016).

SMASH11 specifies that documentation of the functionalities of the system should be made available for the user to access no matter in which part of the system they are located. This can increase efficiency and the user's knowledge of the system and reduce errors (Inostroza et al. 2016).

SMASH12 specifies that the system should provide comfortable physical interactions by adhering to ergonomic standards. For example, one should consider the reach of the user's thumb in the placement of interactable elements. This can increase both ease-of-use and efficiency, and combat exhaustion due to uncomfortable interactions (Inostroza et al. 2016).

2.3.2 Mobile user interfaces for the elderly

Dodd, Athauda and Adam (2017) says that research suggests that the best solution to lessening the economic burden of taking care of the growing elderly population is to enable the elderly to be as self-sufficient as possible. Extensive research has been done to explore solutions to this problem for advancing technologies in the caregiving sector. In order to produce technology accessible to the elderly, their situation and specific issues need to be taken into account in the design of user interfaces.

Dodd, Athauda and Adam (2017) performed a Systematic Literature Review (SLR) to answer which challenges have been identified for elderly users of user interfaces and which solutions that have been proposed to solve these challenges. Dodd, Athauda and Adam (2017) examined a total of 30 studies that matched their search criteria and based on this summarized user interface issues and solutions for the elderly.

With advanced age comes a gradual diminishing of physical capabilities. The most significant physical issue in the context of graphical user interfaces is impaired eyesight. This can make small text hard to read or make it difficult to identify interface elements to such a degree that it affects the understanding of the interface. Old age affects fine motor skills as well, which may lead to difficulty in performing precise movements. Deterioration of hearing, although not as important as the other in the context of user interfaces, may be an issue if the interface uses sound effects (Dodd, Athauda & Adam 2017).

Dodd, Athauda and Adam (2017) states that cognitive issues may arise with age as well. This affects attention, which increases the time taken to analyze and understand the interface. It may also make it difficult to focus on certain content if there are other interface elements competing for attention. Working memory (memory that pertains to the current activity) may worsen, which affects decision making and problem solving. This can result in problems remembering the steps that was taken to complete a task as well as make it difficult to plan and perform more than three actions in succession to complete a task (Dodd, Athauda & Adam 2017).

Cognitive deterioration from aging also affects long-term memory. This worsen the ability to remember actions needed to complete a task and may require relearning the path taken. This may especially become a problem for unintuitive interfaces. If there is considerable time between use or steps taken, this may also lead to problems in remembering what was previously done and what was needed to complete the task (Dodd, Athauda & Adam 2017).

Dodd, Athauda and Adam (2017) states that computer usage and experience is generally lower for elderly people than it is for younger people. They also have a reduced acceptance of emerging and novel technology as well as ignorance or a disbelief of the benefits of technology which serves as an obstacle to gaining further experience. This may lead to several issues, such as unfamiliarity of commonly used terms, symbols and interactions of a GUI which may heighten the difficulty in understanding it. They may have a limited understanding of the back-end processes of information systems if no feedback is given through the GUI. They may have difficulty in using and understanding more complex input elements such as a numerical up-down control or date input control (Dodd, Athauda & Adam 2017).

Dodd, Athauda and Adam (2017) says the foremost issues are connected to interface and control design. As such, the most critical part to designing an interface for the elderly is to make interface controls, such as links and text inputs, as usable and accessible as possible.

Conforming to established standards for text size, color and font is the most common strategy for accessibility according to Dodd, Athauda and Adam (2017). Dodd, Athauda and Adam (2017) refers to a study on optimal button size and spacing in touch screen user interfaces performed by Jin et al. (2007). Jin et al. (2007) concluded that a square button with the size of 19.05 mm was most preferred and optimal for older adults, but if screen space is an issue the minimum preferred size is 16.51 mm. As for spacing in a row of buttons, Jin et al. (2007) reached the conclusion that older adults preferred a spacing of 6.35 mm.

Dodd, Athauda and Adam (2017) states that studies show that more complex touch screen gestures should be avoided due to the increased difficulty for older adults to perform these kind of gestures (for example, the pinch gesture).

To solve issues related to control elements for the elderly, one should make complex controls more intuitive. Such as replacing a date input with a calendar. Visually representing actions for buttons and links using images or icons was shown to be beneficial (Dodd, Athauda & Adam 2017).

Informing and giving the user feedback of what the system is doing upon actions or throughout a process was also shown to be helpful in making older adults gain understanding of how actions affect the system and how the system works. It also alleviated anxiety connected to distrust of the information system. Finally, implementing context sensitive help was shown to be very helpful for older adults to guide them through selections (Dodd, Athauda & Adam 2017).

The second area of solutions deals with the input controls; the device used to interact with the user interface. Research showed that touch input as used in modern mobile devices led to a much better performance for elderly persons in comparison to other input control methods such as mouse and keyboard or eye-gaze input. Voice input was also identified as a good alternative to touch keyboards for inputting text in smartphones, as these keyboards are commonly too densely populated with buttons to accurately use for the elderly (Dodd, Athauda & Adam 2017).

The third solution area relates to the language commonly used in information systems that are not natural for non-established users of these systems. Replacing jargon with plain descriptive language is helpful for people who are unfamiliar. According to research, short and plain messages that communicate options is preferred as elderly persons may be confused if given too long messages or too many options at once. The elderly also preferred to choose default values presented to them over entering their preferred values if they were deemed to be similar. Presenting optimal default options can therefore be helpful (Dodd, Athauda & Adam 2017).

The fourth and last area of solutions deal with cognitive deterioration due to old age. Dodd, Athauda and Adam (2017) suggests performing cognitive evaluations on the user interface with elderly users to improve learnability and understanding of the interface to make it more intuitive.

2.4 PDSA cycle for self-sufficient improvement

The plan-do-study-act cycle (PDSA), also called the plan-do-check-act cycle (PDCA), is a model for continuous improvement. This four-phase process starts with *plan*; in this first phase goals for change and a plan on how to reach those goals are set. In the second phase, *do*, the plan is executed, and the changes tested. During the *study* phase an analysis of the results are performed as well as identification of what was learned. In the final phase, *act*, action is taken on what was learned. If the plan for change didn’t work, then in the next cycle the plan is altered and tested again. If it was successful, the changes are implemented for the next cycle (Tague 2005).

The PDSA cycle is commonly used to implement change and improve quality in the healthcare system (Taylor et al. 2013). But employing this method for personal health improvement is a relatively unexplored area. It has seen use for this purpose in a Swedish program called “Passion för livet” (Passion for life) for improving the health and quality of life for the elderly (Region Jönköpings Län 2017).

Region Jönköpings Län (2017) states that the program is based on the concept of *active aging,* that is made possible through physical, mental and social well-being throughout one’s life. The program aim to empower the participants in taking control of their life and well-being. This is done through a series of group meetings to exchange knowledge and experiences on topics such as natural aging, physical activity, fall-prevention, food and drink, social community, and medicine. In between meetings, the participants utilize the PDSA cycle, referred to as the Wheel of Life (livshjulet) in the context of the program, to incrementally implement small positive changes in health and well-being. In starting the PDSA-cycle, the participant should consider three questions: “What do I want to achieve?”, “How do I know a change is an improvement?”, and “Which changes can lead to an improvement?”.

To explain the need for change, Region Jönköpings län (2017) presents an illustration of the current situation and the desired situation, where the gap between these states need to be filled in by small changes over time. Through the PDSA-cycle, it is tested if the changes has filled the gap to enable the individual to reach the desired situation. If not, further small changes are tested until such is the case.

Region Jönköpings läns (2017) describes the change process in an 8-step circle: my own reality, conviction, clear vision, work processes, result, find new forms, and culture change. Finally, Region Jönköpings län (2017) encourages participants to actively measure their progress to see if a change is an improvement.

2.5 Synthesis of theoretical framework

Bravo et al. (2018) provides many examples of emergent use for m-health systems which informs the possibilities of the technology. Learning from the conclusions of prior research is an important step, and if taking into account the approach from Hassenzahl (2008), it is not enough to simply develop a usable m-health system that has a utility to the user through pragmatic qualities; it has to support the fulfillment of be-goals to be considered to provide a good user experience. Usage of persuasive strategies and the PSD model by Oinas-Kukkonen & Harjumaa (2009) may provide additional support for be-goals that also may promote long-term use. It is also possible for persuasive principles such as cooperation and competition be used in combination with gamification techniques, which Bravo et al. (2018) points out can encourage use of m-health.

To make sure that the system adheres to a high standard of usability for the target audience, it is important to be well-informed of their needs and issues, so it doesn't become a barrier to be-goals (Hassenzahl 2008). As such, the fundamental heuristics for smartphones laid out by Inostroza et al. (2016) ensures that development follows a standard of usability. The summary of issues and solutions presented by Dodd, Athauda and Adam (2017) is vital to address the needs of the elderly when interacting with the GUI. Using the approach to the PDSA-cycle by Region Jönköpings län (2017) provides the underlying goalsetting structure to support the user to take the action (do) necessary to become their desired self (be). See Figure 2 below for an illustration of the proposed framework for m-health systems for the elderly.

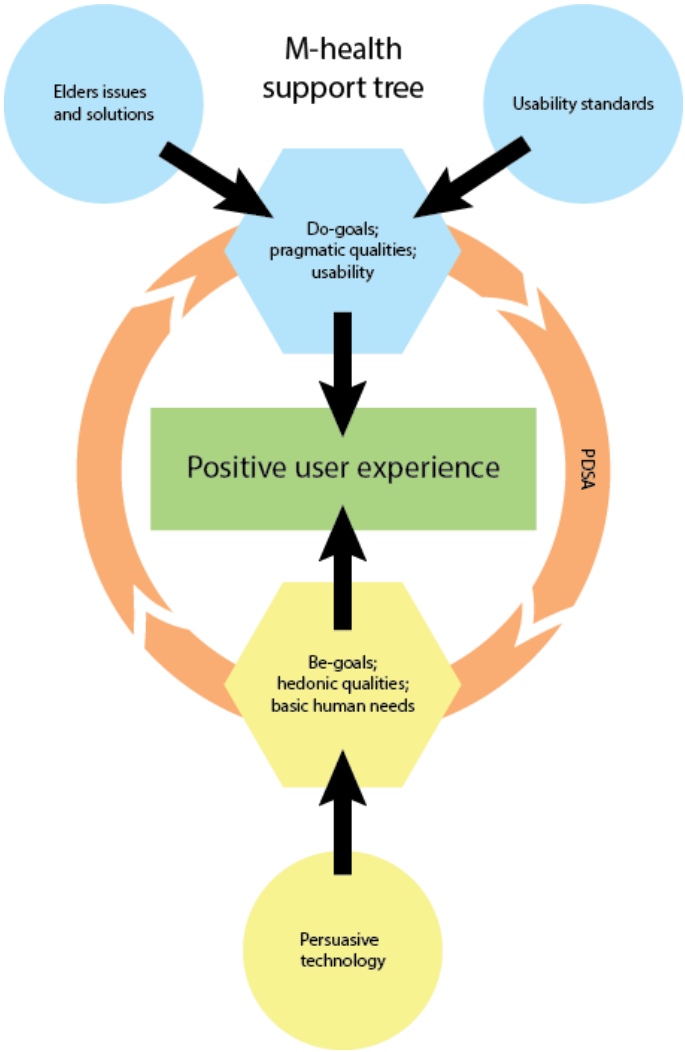


Figure 2 – Proposed framework for m-health to support a positive user experience for the elderly. Informed by work by Bravo et al. (2018), Dodd, Athauda and Adam (2017), Hassenzahl (2008), Inostroza et al. (2016), Oinas-Kukkonen and Harjumaa (2009), and Region Jönköpings län (2017)

3 Methodology

This chapter contains an overview of traditions within case study and qualitative research and defines the researchers overview of the subject followed by the methods used in this study and their detailed execution. After this, a description how to grant levels of validity and reliability of gathered data is presented, as well as measures taken to preserve ethical integrity of the methodology.

This study utilized a case studies research tradition. This design is found in many fields, particularly in evaluation where researchers create an in-depth analysis of a specific case. The case itself had limitations regarding time and activity, and case specific detailed data was collected using different data collection procedures within a sustained time-frame (Creswell 2018).

A qualitative research approach was chosen for the case study. According to Creswell (2018), qualitative research strives to gain a deeper understanding of the social situation, event role, group or interaction. Qualitative research is an investigative process where coding data is collected from the object of the study. This approach was chosen as all data in this case study would be subjective and based on experiences and characteristics of individuals and therefore wholly non-numerical, making a quantitative approach unreasonable in this context.

Presenting an array of characteristics inherent to qualitative research, Creswell (2018) states it is the researchers themselves that has the role of collecting, interpreting and presenting the data. Qualitative processes are emergent for its researchers, and the researcher cannot predict the outcome as some processes may change during the course of the study. This is an indicator of the researcher delving deeper into the subject.

Creswell (2018) says the core concept in qualitative studies is to learn about a problem from the participant's point of view, which in turn gets addressed in the research. Qualitative researchers try to create a bigger, more complex picture of the problem at hand. From the multiple perspectives and factors a situation creates, a bigger picture emerges where several factors interacts in different way.

Additionally, qualitative research inquirers need to reflect over their role in the study and the assumptions they ascribe to the collected data. This not only refers to expressed biases and values in the data of the study itself, but also how the researchers own background could affect the course of the study. As a researcher in qualitative studies, one should strive to leave biases and assumptions outside of the study (Creswell 2018).

Creswell (2018) states that data collection should be performed in a natural setting. This means that one should have face-to-face interactions with the concerned in the

context of the issue or problem being studied. Another characteristic of qualitative data collection is the use of multiple sources of data (triangulation) as to not solely rely on a single data source.

Creswell (2018) claims that data in qualitative research should be analyzed inductively and deductively. Initially, qualitative research is performed inductively where patterns, themes and categories are identified and then sequentially transformed into more abstract units of information. The results from the inductive process can then be deductively analyzed to determine if the data collected is sufficient to support a conclusion or if additional information is required.

As such, even though the case study primarily employed an inductive process, a deductive approach also played an important role in the progression of its analyses. Emphasis was put on identifying the participants meaning of the expressed issue by the researcher.

This case study employed a user-centered design philosophy in the choice and execution of methods. Saffer (2010) states that a user-centered design process focuses on the needs and goals of the users. As such, the users in this case study were directly involved in the design process via participatory methods and design decisions were based on user data.

3.1 Recruitment and participant criteria

Participant criteria was set identically in all methods: (1) elderly or otherwise frail, (2) lives in their own home. This criteria was decided by the research team through information provided through documentation for FRONT-VL and in project meetings with contacts working with the FRONT-VL project. No specific age span has been set, as there are no specific age requirement for being classified as “elderly”. As such, it was determined on a case-by-case basis. The participant is considered to match the criteria for being elderly if: they acknowledge feeling elderly, they are 65 years or older, or they possess a health status that would classify them as frail.

Participants were recruited via convenience sampling through the extended social network of one of the researchers in the case study. Recruitment was carried out in the suburban area of Philadelphia, USA. Potential participants were identified through a match to the criteria for participants. Participants matching the criteria were contacted via telephone and given a brief description of the project and its purpose, and thereafter asked if they were willing to partake in the data collection method in question. After a brief explanation of the method and confirmation of willingness, a date and time was scheduled. General approach to choosing the location of the method was to let the participant decide to assure they were comfortable with the location. When prompted for options, location suggestions include the informant’s house, the facilitator’s house, or a neutral location such as a café.

3.2 Semi-structured interviews

A semi-structured interview is a qualitative interview with a looser structure and higher flexibility than structured interviews which have a more quantitative focus. At its core all qualitative interviews consist of an exchange of dialogue, a narrative approach where the interviewer has certain topics to investigate but with a flexible structure, and a perspective of knowledge to be co-created by dialogue and a reconstruction of the interviewee's narrative and experience; An interview is a social event with learning opportunities for all participants (Edwards & Holland 2013).

3.2.1 Data collection

Firstly, a written informed consent form was formulated that introduced the study as well as the purpose of the interview. Edwards and Holland (2013) points out that being too specific about the purpose of the interview may influence the answers given by the participant. Therefore, care has been taken to speak in general terms about the topic and avoid becoming too specific or detailed in the consent form and when presenting the topic before the interview. Treatment of private information and the rights of the participant were informed in detail in the form as well.

In parallel with the informed consent form, an interview guide was created. interview guide outlines the topics or questions to cover during a qualitative interview (Edwards & Holland 2013). The interview guide was written with a focus on information deemed critical for determining how a persuasive system should be designed from the perspective of the informant in question as well as for determining effective persuasive strategies. The interview guide was also written to include reminders for interview conduct as well as interview structure and flow.

Prior to the start of the interview, the participant was given the informed consent form to read, and thereafter sign to proceed with the interview or decline to abort the interview. Upon completed signing of the informed consent form, an audio recorder was turned on. A mobile application was used for audio recording and creation of .wav audio files. Notes were taken during interviews to help the interviewer keep track of discussed topics.

Questions asked focused on inviting storytelling and dialogue as to encourage the interviewee to answer on their own terms as suggested by Edwards and Holland (2013). Follow-up questions were utilized to encourage deeper explanations and elicit further discussion around a specific topic (Edwards & Holland 2013). Edwards and Holland (2013) also suggest using *verification questions* such as "What do you mean by…" to confirm one's interpretation of the answer. As such, verification questions were used by the interviewer when their interpretation seemed dubious. Interview time ranged from 20 to 45 minutes depending on the participants involvement, or until topics from the interview guide were exhausted.

3.2.2 Analysis

The data analysis process took collected data and transformed it into the same format and subsequently organized the data by clustering and naming similar data, aggregating redundant data, and compared related data as well as unrelated data to find a connection. The formatted data was then transformed using an analysis method suggested by Creswell (2018). A web application was used for the analysis process.

The qualitative data was analyzed using Creswell's (2018) five-step process. The five-step process requires sequential steps to be followed, moving from the specific to the general (the flow of the process can be seen in Figure 3). The process is described by Creswell (2018) as follows:

1. *Organization and preparation of data for analysis.* Initially the interviews were transcribed, and field notes was digitalized. Pictures of the results from the brainstorming sessions in the future workshop was transformed to a digital format as well as notes and supporting guidelines used by the researcher. Transcriptions were done right after the interview was performed to reduce the risk of misinterpret meanings of the answers. Field notes were digitalized, and audio-recordings saved as .wav files. All collected data was stored in the study's data bank.
2. *Read through the data.* Reflections over the meaning of the results from the interviews. Identification of the overall meaning and what the participants have said. Find what credibility the answers have and how will is the information be used. Finding of general ideas and tone as well as overall dept and credibility of the collected data.
3. *Start coding the data. The* process of organizing then takes place where the data gets organized by bracketing it up in chunks and bind it to a category named using actual expression of a participant, in vivo.
4. *Generate a description and themes.* Coded data later acted as the foundation for the different themes or categories as well as description of a scenario, a detailed rendering of information regarding people, places and the events. Creswell (2018) recommends this approach to design detailed descriptions of settings. These themes and categories represent the major finding of the case study, where expressions or statements were lifted by several participants, in a reoccurring theme. It was then applied to a description of use case setting where user and its environment was defined.
5. *Representing the description and themes.* Lastly, a narrative passage where a discussion of findings was conducted where discussion regarding the identified themes was held.

3.3 Future workshop

Vidal (2005) says that *future workshop* is a suitable method for improving the living situation of a target group as well as society as a whole. It is a democratic method that focuses on critique, learning, empowerment and team work.

Vidal (2005) states that a future workshop consists of five different phases: the *preparation phase* (planning the workshop: topic, target audience, location, etc.), the *critique phase* (identifying problems in the topic), the *fantasy phase* (devising utopian solutions to the problems), the *implementation phase* (making utopian solutions realized and implementable) and the *follow-up phase* (workshop evaluation and action plan).

3.3.1 Data collection

The future workshop method was chosen to gain the insight of the target audience in what problems they currently experience and can identify, as well as what solutions they find ideal, in a democratic and participatory approach. The co-ideation platform of a future workshop was considered suitable to involve the potential users of the solution in its design through their expressed concerns and ideas. Thereby the design of the resultant solution is informed by real expressed concerns.

Preparation phase

In the preparation phase, Vidal (2005) recommends setting an unambiguous theme. As such, the theme was first of all set as *elderly health*, with the specific question to be explored in the critique phase defined as: *"What are the issues for the elderly to independently improve their health?"*.

Invitations was initially provided through phone calls. This later transitioned to group text messages to enable planning and provide time slot suggestions as well as general information regarding the future workshop. Participants in the older age bracket was called up to make sure information was reaching out. 6 participants were initially invited, but only 5 could partake during the suggested time slots allowed by the time frame of the case study. As Vidal (2005) indicated that sub-groups within a future workshop should be 4-6 individuals, it was decided that the future workshop should only consist of one group as otherwise the groups would be too small.

Vidal (2005) suggests that the location of the future workshop should be "relaxed and cozy". As per this suggestion, deliberation together with the participants was carried out which resulted in the selected location being one of the participant's home, due to being most geographically available to all the participants and fit the criteria of being relaxed and cozy. A written informed consent form was formulated for the future workshop in the same manner as for the interviews; based on the description by Edwards and Holland (2013). The future workshop was estimated to take 2 and a half hour to 3 hours depending on participant involvement.

Lastly, a “cheat sheet” was created to guide the leader of the workshop. This cheat sheet contained reminders as well as general information of how to conduct the future workshop. The future workshop cheat sheet was based on the same purpose and principles as an interview guide as recommended by Edwards and Holland (2013) to guide an interviewer through a qualitative interview.

Critique phase

The critique phase is the first phase conducted with participants (Vidal 2005). Prior to the official start of the future workshop, food and beverages were provided to the participants. Each participant was provided with a printed copy of the informed consent form and the FRONT-VL project leaflet from CELTIC-NEXT (2016). The project and the future workshop method was briefly explained. The participants were given time to settle in as well as thoroughly read and sign the informed consent form.

When all of the participants had confirmed their consent and signed the form, the workshop officially commenced with the divergent step of the critique phase. The topic was introduced, and the question for this step was written down on a post-it note and placed centrally on a wall. The participants were encouraged to explore the topic and point out any issues they experienced, or they could imagine for this question. As suggested by Vidal (2005), post-it notes were provided for writing down any critique on. The post-it notes were then placed randomly around the post-it for the topic.

Once the issues for the topic was deemed exhausted, the workshop moved on to the convergent phase. The critique post-it notes were discussed and clustered according to their topic and relationship to each other. The clusters were then titled according to its overarching theme. The critique phase took close to two hours to complete. This also included some general talk that not directly contributed to the data bank but helped to reinforce the relaxed setting of the workshop that Vidal (2005) promotes as a factor that improves the quality of the future workshop.

Fantasy phase

After the convergent step of the critique phase was completed, the workshop moved on to the fantasy phase. The participants were informed of the goals of the divergent step of this phase: create a utopia with no restrictions in place (Vidal 2005). To emphasize the restriction-less nature of the fantasy phase, the question “If a genie in a bottle gave you three wishes to solve the problem, what would you wish for?” was asked and continuous efforts were made to reinforce the mindset of having unlimited resources and no laws or technological restrictions to abide by. This was done in order to promote creative and unorthodox solutions and to avoid being weighed down by rational thought processes. Vidal (2005) states that being limited to a wholly realistic context can strangle the creative process and result in that novel and innovative ideas never come to light.

Ideation was conducted using brainstorming techniques; no rejection or criticism could be directed at the ideas and a large amount of ideas were encouraged (Vidal 2005). Ideas were written down on post-it notes and put up on a wall.

When no more ideas could be thought up, the convergent step began. The participants were informed of the purpose of this step; they were told to prioritize and select the most promising ideas that could potentially be implemented if given a more realistic approach and could also solve or minimize the issues presented for the topic.

Implementation phase
As the participants finished up the divergent phase of the fantasy phase, the implementation phase was introduced. After informing the participants of the purpose of this phase, the realizable ideas from the last phase were discussed and refined to the degree that they could be implemented and used in a real-world scenario. With expertise input from the facilitator of what possibilities are available, a strategy to how implementation should be carried out was discussed.

Follow-up phase
Vidal (2005) states that the follow-up phase should wrap up the workshop with an evaluation of the future workshop. As such, the workshop was briefly summarized by going through the critiques, ideas and realizable ideas. The participants were asked if there was any other issues or ideas that had come up, and if they had any thoughts or feedback on how the workshop was conducted.

Vidal (2005) says that in this phase, one should agree to formulate a report that contains the action plan as well as the evaluation of the workshop. This report should then be sent to each participant of the workshop. This structured report on the future workshop was cut and replaced by offering the participants to read the thesis they contributed to when it is published.

3.3.2 Analysis

The analysis of data collected through the future workshop used the same process as the interview analysis: Creswell's (2018) five-step process. Primary difference were that instead of utilizing a web application for the analysis, it was done physically, by hand. This was because of the fact that the data was already written down on post-it notes through the process of the future workshop which made it a more efficient approach.

3.4 Revolutionary rapid prototyping

Overmyer (2002) states that rapid prototyping is a method that is often successfully used in human-computer interaction (HCI) to quickly design user interfaces. There are at least two different approaches to rapid prototyping: revolutionary and evolutionary. In revolutionary prototyping (also referred to as throw-away prototyping) an interactive simulation of the system with limited functionality is created to be used to prototype critical requirements or scenarios that explore vital functions.

End-users can then evaluate and provide feedback if the design meets the requirements. This prototype is then discarded, and the actual system is developed using the findings and feedback from the prototyping phase. Evolutionary prototyping uses a similar approach to user feedback and evaluation but involves developing a software prototype which will gradually develop into the production system instead of being discarded (Overmyer 2002).

Goodwin (2009) states that there are three aspects that usually lead to a good design in short time: developing context scenarios (narrative that describes the system, use context and user) into key path scenarios (scenario that describes needed functions to accomplish the task, grouping of functional elements that are often used together into tool-sets and layouts, and visualizing the system with sketches.

3.4.1 Data collection

For this case study the revolutionary approach to rapid prototyping was used as the method to produce an interactive prototype for the purpose of evaluation by the target group and to serve as a design suggestion for the system. The process used was primarily based on the methods and recommendations of Goodwin (2009) and Saffer (2010).

In an overview, the primarily sequential process consisted of the following methods: *personas*, a design artefact that encapsulates the goals, situation and behavior of users in the target audience (Goodwin 2009); *context scenarios*, an early scenario that demonstrates ideal system behavior in a realistically occurring situation (Goodwin 2009); *key path scenarios*, scenarios that summarize functional needs and elements (Goodwin 2009); *storyboard sketches*, thumbnail sketches of the user interface flow as described by scenarios (Goodwin 2009); *sitemap*, a diagram illustrating the relationship between pages and other information elements (Rosenfeld, Morville & Arango 2015); *wireframes*, a detailed blueprint that accurately displays all the elements for the page with minimal visual design (Saffer 2010); *mockups*, implementation of visual design in the prototype; *interactive prototype*, implementation of interactions and animations in the prototype (Saffer 2010). See below for the complete description of the execution of this process.

Personas
The persona creation process was partly based on the process described by Goodwin (2009, p. 243) and started with identification of variables and patterns pertaining to attitude, behavior, environment and demography. This was done using a process of cross-case analysis of user data (Goodwin 2009) based on the results of the previous interviews. Utilizing Goodwin's (2009) method, the variables structured as conflicting continuum variables such as "younger" versus "older", or "lacking social network" versus "strong social network". The conflicting variable states on the continuum were then sorted into two archetypes with consideration of which interviewee uttered the statement as well as with thought taken to environmental and demographic variables affecting attitude or behavioral variables.

Due to time constraints, the variables weren't converted into a narrative as recommended by Goodwin (2009). They were instead made into a list of persona characteristics. Functional needs were added, the persona was given a name, then a picture representing the persona was selected. See Appendix B for the personas.

Context scenarios
It was decided that a minimum of one context scenario per persona (see Appendix B) should be created, using the persona as the actor in the scenario. The development process of the context scenarios began with formulating a rough outline of the narrative, based on the activities, functional needs and goals of the persona. Inclusion of functional requirements (see Appendix A, Table A.2) in the scenario was also considered were appropriate according to the persona. The narrative was further fleshed out based on the attitude and behavior of the persona. Regard was taken to crafting an engaging, informative and believable story, with a realistic triggering event and an ideal conclusion of the story.

As per the suggestion of Goodwin (2009), care was taken to keep descriptions of system interactions at a high level. Goodwin (2009) states that a solution-agnostic approach for context scenarios is important as locking oneself into specific solutions too early can eliminate proper exploration of design alternatives. See Appendix C for the context scenarios.

Key path scenarios
The process of creating key path scenarios began by copying out passages from the context scenario for each persona that described system interaction. These excerpts were then analyzed to find descriptions of system behavior, functions and interface elements. As per the description of Goodwin (2009) for this method, system behavior and functions were categorized under "functional needs" and functional interface elements were categorized under "functional elements" and described at a high-level. See Appendix D for the key path scenarios.

Storyboard sketches
To move from understanding users to actually solving their problems, Goodwin (2009) suggests storyboarding scenarios to visualize the interface and interaction flow. The storyboarding process was performed by drawing rough thumbnail sketches of each screen and its major functional elements involved in the scenarios as recommended in the scenario-first storyboarding method by Goodwin (2009). Appropriate persuasive design principles (Oinas-Kukkonen and Harjumaa 2009) were considered based on the needs and personality of the personas. See Appendix E for all the storyboard sketches.

Sitemap
In order to clarify the navigational structure and the organization of the information architecture of the system, a sitemap was created. According to Rosenfeld, Morville & Arango (2015), creating a sitemap is useful to evaluate if the imagined architecture is viable in practice. The sitemap was revised as needed in parallel with the storyboard sketches and was made to include all pages of the prototype and their relationship to each other. See Appendix F for the sitemap.

Wireframes
The layout of interface elements as created in the sketching phase were more precisely specified to a pixel-perfect degree and tested according to visual hierarchy and look-and-feel. This was done using prototyping software. As recommended by Jin et al. (2007), the touch-surface of 19.05 mm (72px for 96 DPI displays) squares were implemented for interactable elements. The lack of surface recommendations for rectangle shaped surfaces led to testing and resulted in touch surfaces with y-axis at the minimum recommended square size of 16.51mm.

The wireframes were tested on a smartphone to make physical measurements for interface elements. Following the recommendation by Saffer (2010), dummy text and images (placeholders) were used to reserve the space for content yet to be created. Elements intended to be functional were annotated and had its intended behavior described beside the interface wireframe, as suggested by Saffer (2010).

The wireframes went through several iterations, based on testing different implementation solutions for SMASH (Inostroza et al. 2016) and interface solutions for the elderly as presented by Dodd et al. (2017), that were not taken into account in great detail during the sketching phase. Because of time constraints, wireframes focused on capturing the layout of screen types rather than making a wireframe for all screens of the prototype. See Appendix G for all the wireframes.

Mockups
Implementation of visual design, graphics and color were tested on screens from the wireframes in accordance with the method described by Goodwin (2009). Goodwin (2009) states one should produce a design specification that defines the rules for the

visual design. Based on this, a design specification was produced in parallel with the visual design testing. It was produced through first creating a canvas with design elements and real-world objects that captured the look-and-feel of the desired visual design, similar to a mood board that Arvola (2014) advocates using to define the look-and-feel of a visual design.

The design specification was then created through information gathered through visual design testing and the mood board. The design specification included guidelines for: font (size, weight style), color scheme, visual design guidelines (rules for size, spacing, border radius and shadows).

Shadows, with defined offset, blur and spread, were used to increase the affordance of interactable elements. Material Design was chosen as the underlying design style due to its widespread use in mobile interfaces, which may aid efficient use due to recognition from using other applications through SMASH4 described by Inostroza et al. (2016). A web-based contrast checker was used to assure that the visual design followed contrast and legibility standards for font and background elements as recommended by Dodd et al. (2017). The contrast checker was set to validate according to a contrast ratio of 7:1 as specified by the enhanced contrast criteria set by W3C (2018).

Interactive prototype

Saffer (2010) states that when most of the details of the refinement process has been established, it is time to move away from static screens and begin to incorporate interactions and animations. Using functionality from prototyping software, the mockups were remade into an interactive prototype. Fluid button animations and page transitions were added to promote aesthetics and user experience while not negatively impacting efficiency of use as specified by SMASH8 (Inostroza et al. 2016). The additional feedback from interactions provides a deeper understanding of the system and was implemented to enhance the mapping to the real-world.

3.4.2 Analysis

To keep in compliance with SMASH (Inostroza et al. 2016) and elderly issues and solutions (Dodd et al. 2017), an analysis utilizing a heuristic evaluation was done. The evaluation was performed on the interactive prototype to gauge the adherence of the implemented solutions to the heuristics.

3.5 Usability evaluation

A usability evaluation is a test where participants perform a series of realistic tasks using the artefact to discover usability issues (Goodwin 2009).

3.5.1 Data collection

The process for the usability evaluation began with identifying what needed to be evaluated, as recommended by the evaluation method by Goodwin (2009). The user evaluation strived to investigate usability and the experience of the implemented design to see how it holds up in scenarios in a simulated context. It also focused on investigating if there was any aspects of the design that wasn't considered, or other problems not discovered in the analysis of the prototype. According to Goodwin (2009), an evaluation can serve several purposes, including, but not limited to improving design and demonstrating the effectiveness of the design, which was the purpose of this evaluation. The findings of this method thereby serve to inspire eventual revisions of the prototype to improve the design.

Firstly, an informed consent form specifically for the usability evaluation was formulated. As suggested by Goodwin (2009), several tasks was formulated for the participant to perform to evaluate the prototype. Care was taken to follow the advice given by Goodwin (2009) that an effective task is based on user goals, is related to important issues for success, has an appropriate scope and a limited and predictable set of solutions.

During the execution of the method with the participant, a test smartphone was used which had the prototype preloaded. The participant was first introduced to the method and project. Thereafter the participant was given the informed consent form to read at their pace. After the participant had signed the form, the participant were given the test smartphone and the first task verbally. Next task was given upon completion of the previous one.

The evaluation used the think-aloud method as suggested by Goodwin (2009); participants were encouraged to continuously articulate their thought process aloud when using the prototype. In cases where the participant got stuck or could not understand the instructions, further hints were given without explicitly giving them the solution to the task. In moments when the participants seemed unknowing about what to do, they were reminded to verbalize their thought process. Notes were also taken as a complement to the think-aloud method. In two cases video recording was used with a separate smartphone in order to capture interactions for later analysis.

After the tasks were performed, a brief discussion followed where the participant were given open questions regarding the current design, potential improvement and given the opportunity to give feedback about their experience with the prototype.

3.5.2 Analysis

The analysis of data collected through the usability evaluation used a similar process as described in the interview analysis in Creswell's (2018) five-step process for analyzing qualitative data. Key difference was that potential problem areas were identified along with finding suggestions for improvement.

3.5.3 Prototype revisions

Following the analysis of the results gathered through the usability evaluation, the prototype was revised with consideration to the feedback submitted by the participants in the usability evaluation as well as through any discoveries made through observation of the participants' use of the prototype. In addition, the visual fidelity of the prototype was improved. All of the prototype screens were peer-reviewed and analyzed according to set standard and other potential improvements by the researcher not involved in the creation of the prototype screen in question.

3.6 Validity and reliability

Validity in qualitative research does not carry the same connotations when compared to quantitative — neither does reliability or generalizability. This is due to the layers of interpretation and the unique nature of qualitative data. Researchers needs to take active measures and convey the steps performed to check for the accuracy and credibility in their findings (Creswell 2018). To ensure qualitative validity, several strategies, as suggested by Creswell (2018), was employed to assess the accuracy of the findings. Gathered data was triangulated. Data was collected from interviews, a future workshop and a usability evaluations and was converged jointly into themes.

Extensive efforts was made to provide detailed descriptions of settings for conducted data collections as well as being as transparent as possible; negative and inconsistent information was intended to be presented as accurate as possible. Lastly, having a supervisor throughout the study that has been asking questions and reviewing the project, making the account resonate with people other than the researchers, involving interpretation beyond the researchers, adds to the validity. The peer debriefing was also taken into account between the researchers in the study and cross-reviews were held internally (Creswell 2018).

In efforts to have a reliable approach in this study a thorough documentation of the different procedures has been made. Also, the quality of the documented data was controlled to avoid obvious mistakes, such as during transcription of interviews. Results, when possible, were crosschecked against previous research where codes have derived independently to ensure higher accuracy of the codes developed in this research (Creswell 2018).

3.7 Ethical considerations

Creswell (2018) underlines the importance of ethical considerations in qualitative research. The researcher has an obligation to respect needs, rights, desires and values of the participant. Also, use and storage of personal data should not extend the need of the study. To follow this, a signed pledge was made through a written informed consent form to delete all recordings made of the informant.

Edwards and Holland (2013) states that gaining the informed consent of the participant is ethical good practice. As such, all data collection methods involving participants required the signing of a written informed consent form that was custom made for the method in question. The informed consent form contained information about: the rights of the participant, the project, purpose and conduct of the method, and the treatment of data and personal information.

4 Results

This chapter presents the empirical findings obtained through data collection and analysis.

4.1 Semi-structured interviews

Four interviews were conducted. The participants were 67 to 89 years old and were living at home in north east USA. Three of the participants were women and one was a man. Two participants were retired. Two participants lived alone, and two of them lived with a spouse. The interviews took place in the homes of the participants and lasted between 20-45 minutes.

The interviews primarily revolved around the topics: attitude towards personal health management, attitude to technology and technological self-efficacy, usage of assistive tools for personal health, self-improvement behavior and the participants attitude and behavior towards goal-oriented processes (motivating and demotivating factors). Findings are presented below.

4.1.1 Attitude towards personal health management

All of the participants stated that health is an important part of their life and that they took active measures toward maintaining or improving personal health. The participants considered daily activities sufficient for a healthy lifestyle and as a substitute to instructed workouts. The participants also embraced their everyday activities (such as cleaning, yardwork, walking the dog, playing with grandchildren and moving around at workplaces) to promote healthy living. Activities promoting mental health and well-being included reading, entertainment, and one participant expressed playing games on the computer.

Motivating and Demoting factors

A common opinion for motivation was the sensation of well-being and the social surplus it generated when done in groups or with an instructor. Relationships with people that had the same mind-set and purpose, and instructors was stated as a determining factor for participation in organized workouts. Also, the feeling of self-accomplishment was stated by one participant as motivator.

Scheduling or knowing beforehand about upcoming health-related events was making it easier to partake and the consistency of workouts was stated to be essential in upholding a regular workout routine. The competitive aspect was a reoccurring topic; when asked about motivating factors one informant answered: *”... because on our Fitbit we've had team challenges we, you know, try to beat each other, so yeah that”*. Competition was considered something that would act as a motivator and also have the potential of increasing the intensity of a workout session.

The times when the workout did not happen it was normally claimed to be due to time restriction or “just life”. In one case the informant stated that the television was a distraction that caused exercise to be neglected. In another case, the informant simply stated that workouts were not needed at the time.

4.1.2 Attitude towards technologies

When it comes to relationship with technologies the attitude was polarized. A majority didn’t foresee it bringing any benefits in their daily lives and was currently not using any m-health tools. In one case the participant had a strong opinion against technologies and its effects on society. Another expressed it was helpful as well as expressed an enjoyment of using “the latest gadgets” and saw several benefits in everyday life using these gadgets.

Usage of technologies in daily life

Smartphones were used by all participants. Motivation was normally based on the social possibilities and for relatives to get in touch. The primary usage of the phone was talking and texting among older participants in the age group and the participants in the younger bracket used applications on a daily basis. One of the older participants used Uber (taxi service application) as a primary way of transportation but stated that was the only app she was using. The other person in the elderly bracket connected her hearing aids to the phone, where settings such as volume and noise reduction were managed. This functionality has never been used by the participant and remains the same as it did when it was initially set up.

Usage of assistive tools for personal health

One participant found great benefits of Fitbit (smartwatch and mobile application for activity tracking) and the competitive aspect of the related software. The same participant mentioned the belief in the benefits of the different health related tracking devices and were also tracking sleep patterns. The other participants did not use m-health tools today. Another participant, that is living alone, was using a necklace with the ability to set off an alarm and get assistance in case of a fall injury at home.

Technological self-efficacy

Participants explained how technology worked against them and the society as a whole as well as their inability of achieving intended goals when working with technologies.

Learning approach

All participants were relying on someone close to them to help them with usage and orientation of their smartphones. Primarily a relative, but in some situations a neighbor. Two of the participants were entirely reliant on help from an external source when needed to be introduced to a new application or technology. The other two used a trial-and-error approach and relied on their knowledge from prior

experiences but stated that they would turn to a younger person when in need: *"I can normally figure out how to use a lot of it… or I will ask a younger person"*.

Usability
The two oldest participants expressed a problem with user-input, especially using the keyboard. One participant used a stylus for the keyboard: *"Well texting, I can do that with a stylus but a can't do it with my hands"*. The other older participant refrained from texting all together and was primarily using the phone for voice calls. The same participant also encourages people that want to get in contact to use the landline when calling since it was easier to hear.

4.1.3 Self-improvement behavior

No participant used any structured measures for self-improvement, except one participant mentioning weekly pilates as part of measures to improve well-being.

Goalsetting
The goals stated were normally short-term goals and primarily revolved around completing daily tasks. Staying healthy and "keep moving" were goals among the participants. Measures taken to achieve this was by incorporating activity into daily tasks. Task-oriented processes was a common topic of discussion in the form of to-do lists, but participants could not further specify their motivation for performing the tasks.

4.2 Future workshop

The future workshop was carried out in Philadelphia, USA, on Saturday May 11th, 2019. The total length of the workshop was around 3 hours and 30 minutes and included 4 participants and 1 facilitator. The participants were between the age of 65 and 87. Two of the participants were women and two were men. The results from the future workshop below are divided into the three phases that are conducted with participants.

4.2.1 Critique Phase

Critique points was divided into the following categories regarding the problem: social, society, mental, physical, healthcare, cultural and living situation.

Social
One of the most discussed areas revolved around the social topic. All participants agreed that the social aspect is a root cause of the problem. Friends as well as family was often the main motivation to take action in personal health. In older age, friends and families becomes more spread out and there are few to no known outlets to tie new relationship bonds and companionship. The participants believed this issue would be even more severe in cases where the person is living alone than if they had a spouse or significant other. Noted is that the social aspects were considered more important than accessibility to workout equipment and other health improving tools.

Society
Political influence was lifted as a critique point and discussed thoroughly. Current laws and priorities were not inclusive of the elderly population and have not adjusted properly to the shift in proportions among the population where the elderly and, as such, retired population is bigger which puts an increased economic pressure on society. Also, income and social security in this population was not enough to make a decent living once retired. It was stated that it was insufficient to cover medical costs and even less to afford the costly options of preventative actions.

Mental
The participants expressed a stigmatic feeling around mental health issues among the elderly population; elderly were often neglected when it comes to depression. Health issues and fading social relationships was believed to be contributing aspects to the problem. Further, the participants agreed on perceiving an increase of laziness and supposed it could be related to the changes of society where reasons for manual labor have diminished or might even be related to the mental issues related to ageing.

The participants also believed that there was a lack of motivation to get regular health checks and believed this was tied to mental issues.

Cultural
Lack of religion was lifted as a critique point; not having God (or spirituality) in one's life was suggested to be a reason for lack of self-motivation to improve one's personal health as well as a cause of depression. Another cultural aspect contingent to the problem lifted was the impact of the television. Too much time and endless options minimizes spontaneous activity and, as such, contributes to the problem of getting involved in their personal health.

Physical
Dwindling physical health was lifted as part of the problem. If physical health reaches a point were preventative measures no longer is possible it becomes a downwards spiral. As a result, physical (and mental) health takes a toll.

Healthcare
Current healthcare situation was considered inadequate. Lack of trust in doctors and health care personnel caused their recommendations to be ignored. An inability of conveying information to the elderly in regard of the problem at hand was believed to promote the lack of involvement in personal health due to lack of conveying the importance and what active measure that they could personally take.

Living Situation
Houses are often built with the intentions of having families living in them. At later stages in life the size of the house can be a problem, hard to maintain, costly and comes with architectural decisions that can cause potential harm in regard to fall

related injuries in older age. These were believed to be adding to the problem. The topic of being “home trapped” where it didn’t make financial sense to move and the high cost of alternative living being too high was mentioned too.

4.2.2 Fantasy Phase

In the fantasy phase, the utopian solutions to the expressed issues were reduced until realizable. These realized solutions include: elderly communities, social-oriented workouts, geriatric medicine, and promotion of health beneficial activities.

Elderly communities

One expressed solution was elderly communities with house designed according to their needs, such as: no bath tubs, addition of railings throughout the house, safety timers for hazardous appliances, one-story houses (no stairs). Emphasis should be on the simplicity of maintenance so house work such as gardening should be optional but embraced if a personal botanical interest exist. All necessities for high quality living should be within walking distance. Such as grocery store, pharmacy, doctor’s office/hospital, social areas such as bar and restaurants. The option of automating the process of grocery and pharmacy shopping should also be available to get home delivery or pick-up.

Exclusion of cars is preferable and there is a belief that having no television inside the houses would create a better community. The weather of the community need to be considered and should thus be placed closer to the equator. Winter affect the living conditions and makes it harder to move around due to the cold weather and snow. Snow and ice restricts mobility and also increases the risks of fall-related injuries. The darkness associated with wintertime was also believed to be contributing to potential depression.

It should be an inclusive community where democratic decision making is done by the residents. Meaningful professions can be exercised, but the main motivation shouldn’t be monetary. The community should have a flat hierarchy as to create better relationships among the residents. The population and size of the community should be just enough to have some type of anonymity but still feel that is inclusive of all its residents. Though, no exact number was defined.

Moving in at a younger age, if no reason or will to stay at home was present, was believed to be better in order to make it possible to live longer independently. It should be a part of life to look forward to, no stigma of moving to such a community as they perceived there was in moving to a retirement home.

Living under such conditions creates a platform where social involvement and reasons to take active measure of improving personal health are very credible according to the group. The community is practically reducing the demotivating factors and hinders to giving its members any reason not to get involved in their

health as an indirect, natural consequence of living under such conditions. Whether it is conscious or not.

Social-oriented workouts
The feeling of being a patient is not considered to help the reasons of involvement in personal health. It was expressed that there was a feeling of being a burden for the instructor/doctor/therapist, as well as a fear of doing something wrong. The group believed that the social aspect is a much bigger motivator, especially when it comes to being involved over time. One should be able to partake in an uncompromising manner; no musts, just to be there because of enjoyment. To achieve this, not only does the instructors/therapists have to be knowledgeable in their subject but also have a great share of social skills and be able to inspire the participants to get involved using a social and close-together approach.

The group expressed there was no need for it to be a gym setting with machines and free weights, neither is there any need for it to cater to a solely elderly demographic. But it was believed that a common goal in the group would be beneficiary. An activator/trigger to get involved was also believed to be beneficial. Not a fear mongering approach by warning about implications if you are not partaking but rather a motivational and positive one. This trigger could be an instructor informing them personally, and stated by the oldest participant, someone (preferably someone close) to bring them along.

Fun-minded competition was stated as a great motivator and something that inspired increased intensity of workouts. To include small competitive aspects in different parts of the session was pointed out as helping the cause.

Geriatric medicine
Discussions after the brainstorm suggested a push in development of geriatric medicines, especially in areas where preventative measures are not known or possible and affects quality of life. Such as dementia, dystrophia and Parkinson's disease. Such diseases could greatly reduce involvement in personal health and the independency of the affected.

There is a resistance against the magic pill mentality in the group. Medicine is not the solution to most problems, and a distrust in doctors' ability to prescribe medicine in an appropriate manner is stated, but in certain cases it is the only solution and it is then a solution to be grateful for. The distrust is founded in the personal incitements among doctors based on relationship with medical corporations decides what medicine is being prescribed. It is still believed that an involvement in one's personal health can avoid medicine reliance.

Promotion of health beneficial activities
It was expressed that one solution could be subventions and promotions at a government level, where personal financial status should not be a hinder for elderly to

get involved with their personal health. Thus, shifting priorities in social security and embrace involvement. No punishment for people that was not getting involved in their personal health was suggested but reasons other than personal motivation itself should be diminished. Meaningful incitements should be provided, such as rewards or positive feedback, or promotion of healthy living. Discussion around if reward system with money could have socioeconomic benefits was held with the conclusion that it potentially would.

4.2.3 Implementation Phase

In this phase the implementation options for the previously realizable solutions were discussed, including: elderly communities, social-oriented workouts, geriatric medicine, and promotion of health beneficial activities.

Elderly communities

To be able to implement this, it requires reaching out to today's developers of elderly communities and have a discussion on how to fundamentally think in regard to layout and purpose. Also, what group that is being targeted to live in the community need a re-haul. One problem that needs to be solved is the disconnection with the current home and how personal belongings are being handled. Streamlined credited contributions to the community itself where excessive belongings is either used by the community or provides help to redistribute it or sell it.

In the U.S. today, living in communities intended for the elderly are too expensive for a majority of the population. Suggestions to solve this problem is to have the ability to work for the community itself, but with workdays designed with an elderly person in mind with shorter days and less physical and psychological stress. It was agreed upon that the lack of work that respects age is unsatisfactory today.

Social-oriented workouts

The group suggests that the market have not been able to reach its demand for this category in the context of elderly. Potentially due to a belief in that there isn't any demand for such a service. Suggested actions could be to find ways of shining a light on the concept through a petition or other means of getting a collective voice heard.

Geriatric medicine

The participants says that state funding is required to implement further development of medicine for the elderly.

Promotion of health beneficial activities

The participants state that forces from a higher instance is required to make this idea into a reality. Research need to be done to determine potential socioeconomic benefits of increased government funding.

4.3 Revolutionary rapid prototyping

The end result of the rapid prototyping process was an interactive prototype of an m-health application supported by persuasive technology focused on social aspects to promote usage by the elderly. See Appendix H for all screens of the rapid prototype.

4.3.1 Persuasive design principles

The main (persuasive) feature of the proposed application was an elderly community. This was inspired by the elderly community solution given by the participants in the future workshop, as well as interview and workshop data that suggested that the main motivator for healthy behavior relates to social aspects. Additionally, loneliness was perceived as a significant demotivator for healthy behavior and also exasperated negative cycles of mental and physical issues.

The community page, as can be seen in Figure 3, is intended to facilitate communication between elders and leverage persuasive design principles such as: *social learning*, *social comparison*, *normative influence*, and *social facilitation* (Oinas-Kukkonen & Harjumaa 2009). The data also suggest that fun-minded competition is a prime motivator for healthy behavior, thus a challenge function was implemented based on the *competition* persuasive design principle (Oinas-Kukkonen & Harjumaa 2009). Using the challenge function, one should be able to send challenges to other users for specific types of exercises with certain set goals to win.

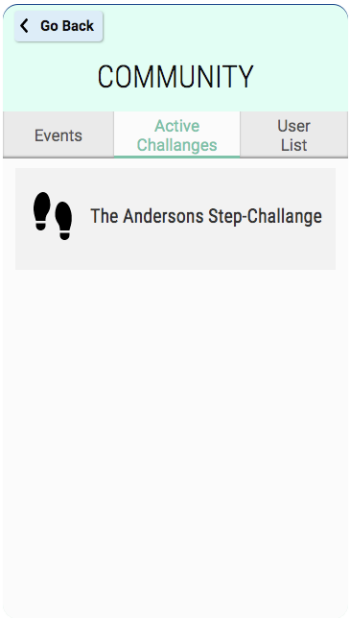


Figure 3 – Community screen: challenges

As can be seen in Figure 4, the proposed application should reward the user custom achievement ribbons based on reached goals that can be displayed publicly. According to Oinas-Kukkonen and Harjumaa (2009), this can have great persuasive power as can be seen through the *rewards* design principle.

One feature that wasn't directly demonstrated through the rapid prototype due to time-constraints, but were a requested feature through the context and key path scenarios (see Appendix G and Appendix H) and included in the storyboard sketches (see Appendix E, Figure E.7 for example implementation) was a notification system that reminded the user of daily goals that was intended to leverage the *reminders* design principle. Additionally, the notifications were intended to provide praise (using the *praise* design principle) when goals had been reached, when a certain amount of progress had been made towards a goal, or upon setting a new personal highscore. This feature can be seen in the storyboard sketches in Appendix E, Figure E.7 and Figure E.9.

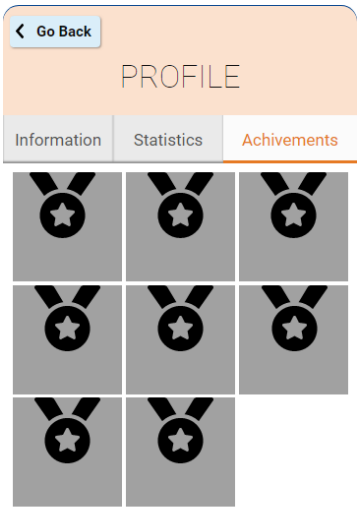


Figure 4 – Profile: achievements

The proposed application is intended to use a goal setting system based on PDSA. After choosing type and condition for the goal, a plan shall be automatically generated based on the user's current average for the type. Then this plan shall make an incremental improvement over time toward this goal. This also leverages the *suggestion* design principle that says that suggestions will have great persuasive powers (Oinas-Kukkonen & Harjumaa 2009). As this plan is made specifically for the user, it also uses the *personalization* design principle. See Figure 5 for the plan

screen. The creation of goals employ a step-by-step wizard (see Appendix H, Figure H.14–16) that implements the design principles of *reduction* and *tunneling*. The goal system itself is also an implementation of the *reduction* principle, to simplify the process of performing a target behavior.

Figure 5 – Goal: plan

The proposed application also intended to make use of the *self-monitoring* design principle. Oinas-Kukkonen and Harjumaa (2009) says that a system that tracks a users performance support a user in achieving their goals. This self-monitoring function comes in the form of user data that is visualized through goal plan statistics, profile statistics, highscores, achievements and notifications.

4.3.2 Information architecture

Interaction options and information for each screen were kept to a minimum to reduce the occurrence of cognitive overload and time needed to analyze the information presented on the screen as recommended by Dodd, Athauda and Adam (2017). At most, the screen presents six interactable elements, as can be seen on the home screen (Appendix H, Figure H.6).

As Dodd, Athauda and Adam (2017) bring forth memory issues as a problem for older adults, a shallow two-level hierarchy was used to reduce the user's memory load in navigating the application. Each section in the second level used tabs and/or modal windows to present additional information without navigating to other screens or deeper in the hierarchy. This also enables the user to easily navigate back to the home screen by the ”Go Back” button in the interface.

4.3.3 Design specification and visual design

The visual design of the prototype aimed at being simplistic and minimalist to combat information overload as suggested by SMASH9 (Inostroza et al. 2016). The design also strived to incorporate affordance enhancing elements via use of shadows and input element design inspired by real-world equivalent objects as recommended by SMASH2 (Inostroza et al. 2016).

Font

Roboto was used for paragraphs and Roboto Condensed for headers. The font size for body text is 17px with a line height of 1.2 was set to ensure good readability. A contrast ratio of a minimum of 7:1 for the font was implemented according to the enhanced contrast criteria by W3C (2018). See Appendix H, Figure H.1 for font guidelines from the design specification.

Size and spacing

The button size of the home screen navigation tiles was set to 156 by 124 pixels with 24 pixels between the elements, in accordance with the standard suggested by Jin et al. (2007). Buttons were kept to a similar touch surface as the tiles. Input fields (such as checkboxes and toggle switches) were also made bigger to increase the touch surface.

Color

The color palette was primarily based on the colors used in a selection of different colored logotypes acquired from FRONT-VL. Color was used to introduce a separation between different sections of the interface. As seen in the home screen (Appendix H, Figure H.6), each button tile has an assigned color. The page navigated to through the button tile inherits the same color scheme as that of the tile. This was done to add further clues to identify where in the system the user is located.
See Appendix H, Figure H.2 for color guidelines from the design specification.

Icons and images

Icons were fetched for free from Font Awesome (2019). Profile image of the Alice persona (see Figure 8 for usage in the prototype or Appendix B, Figure B.1 for the full image) obtained for free from Pixabay (2019).

4.4 Usability evaluation

Four participants between the ages of 69 to 89 took part in the usability evaluation. Two of the participants were women and two were men. One of the participants had also participated in one of the prior interviews. All of the participants owned a smartphone but had varying experience and only used a few different applications.

The test was performed using a Google Pixel 2 XL. The test phone had 6-inch (150 mm) P-OLED display panel, 2:1 aspect ratio and a 2880×1440 resolution at 538 DPI. The physical nature of the test phone is bigger than average, but the actual real estate

of the prototype was held to 360 × 640 pixels. The actual size of the test-interface was close to average phone size. Each evaluation lasted between 15 and 20 minutes.

Initial Impressions
Two participants expressed a feeling of being able to perform the tasks when presented with the prototype, pointing out its simplicity. *"This looks pretty straightforward"*, was stated by a participant after glancing over the home screen.

Flow
No participant expressed being lost (or seemed lost) or that they didn't know where they were in the hierarchy of the interface. They expressed no problems navigating back to a previous screen and did so without second thought. No participant commented on the use of modals and observations indicated that there were no difficulty in understanding how modals work. This also applied to the goal setting wizard that used modal windows.

Usage
Two hands were used across the board; the device was held in the left hand and the right index finger was used to interact with the interface. Usage of the built-in on-screen keyboard caused issues leading to erasing characters, and in one case the typos were left as-is. The movement of the on-screen content when the keyboard folds out caused confusion in another case. When any form input was required the older (89-year-old) participant took longer time in comparison to the younger participants.

Problem areas
In one case the tabs were not immediately understood, but after figuring out how they work the participant did not have any further issues with them. Same participant were quick to perform touch gestures and only glanced over the sub-page for a few seconds before navigating back to the home page. This participant got confused by the low fidelity sections of the prototype where placeholder text occurs, and seemingly functional elements were non-functional. The participant also failed to find the tab *Achievements* under the profile page when the profile page was visited.

The other three participants had a more methodical approach and read through all the options and systematically looked through their options. Then, if what they were looking for was not found on the sub-page they navigated back to the home screen. In two of the discussions following the evaluation it was stated that the tabs did not have a real-world feeling; the tabs didn't look like what they imagined directories to look like — but its functionality was still understood.

In the first two evaluations the participants ran into problems when logging in when the onscreen keyboard covered the log-in button. The prototype was adjusted to resolve this issue for the latter two evaluations.

In one case the system's back button was used instead of the back button in the application. This navigated the prototype back to the welcome screen and not the home screen as expected by the user. Though, this is an issue pertaining to navigation of the link given by the prototyping software rather than an interface design issue. One informant also didn't immediately understand the on/off slider that can be seen on the settings page (Appendix H, Figure H.7).

The task that asked the participant to find current challenges caused some problem with connecting the *community* page to challenges. All participants took additional time when reasoning over their choices on the home screen; two of the participants navigated to the profile page, while the other two navigated to the community page.

4.4.1 Revised prototype

The revised prototype focused on implementing changes according to feedback obtained through the usability evaluation, as well as adding demonstrations of functionality missing from the rapid prototype. The fidelity of the prototype was also increased. The revised prototype added icons from Mozilla (2019), used under the license CC BY 4.0 (Creative Commons 2019). See Appendix I for all the screens of the revised prototype.

An array of changes were implemented. Participants expressed the tabs in the rapid prototype didn't have a real-world feeling, thus the tabs were made more alike to real-life folder tabs. Help text was added to each home screen button to easier understand the contents of each section navigated to by the button. The help section on the home screen was removed and replaced with a contextual help button on each screen. The goal setting wizard were reduced to two steps as the third step was deemed superfluous and merely added interactions required to set a goal.

Screens were added that demonstrated notifications for goals and challenges and implementation of the *praise* and *reminder* design principles (Oinas-Kukkonen & Harjumaa 2009). See Appendix I, Figure I.13 and I.14.

5 Discussion

This chapter presents discussions and reflections regarding the findings of the research and how that correlate to the research question, as well as reflections on the chosen research methodology.

The research question in this case study asked: *How can persuasive technology promote usage of an m-health application by the elderly?*

From our findings, there are many arguments for including persuasive strategies in m-health systems. We find that persuasive technology can play a vital role in promoting the usage of m-health systems and the adoption of behaviors and attitudes beneficial to the health of the elderly.

Our findings, based on results from four interviews and a future workshop, is that the primary motivation for healthy behavior among the elderly is derived from social aspects. Informants expressed that social aspects, such as peer pressure, competition or social belonging, is the driving force behind long-term self-improvement. In cases of diminished motivation to keep to healthy behavior, a lack of social value of this behavior was cited as a prevalent reason.

Furthermore, health concerns or healthy behavior was not seen as enough of a reason of its own to take action in improving personal health. The importance of taking care of oneself is seemingly already profoundly known among elderly, as is the will to be healthy, but active measures are still not widely taken. Therefore, merely informing about the importance or the consequences of not taking active measures of improving one's personal health is not enough. A more positive approach is warranted.

Healthy behavior needs further support from social aspects, or other aspects important for the individual in question. This implies it is not viable to simply create an m-health application with solely pragmatic qualities. The findings seem to support the stance taken by Hassenzahl (2008), that to properly satisfy the user and provide a good user experience, the system need to offer ways for the user to fulfill be-goals. From the findings, the most prevalent be-goals could be assumed to be socially inclined in this context. For example, "being liked by their peers", "being part of a social group", "being respected", or similar.

From a persuasive standpoint, these findings suggest that a persuasive strategy leveraging social support design principles would be the most effective at causing a change in behavior and/or attitude in the elderly.

One social aspect issue that has an impact on motivation for self-improvement is in the relationship between the elderly and caregivers or trainers. The informants in this study perceived that oftentimes the elderly is merely seen as a patient, as someone

that need care. This is interpreted as the elder being a burden and looked down upon rather than being seen as a person on an even-level, which in turn demotivates the elder in participating in health activities or take action to improve their health.

This situation could be improved with the use of the *social learning* design principle. Using this principle, the elder can learn from their peers and adopt their behavior (Oinas-Kukkonen & Harjumaa 2009) instead of learning from a perceived authority figure. In a system incorporating social learning, there would also be an opportunity for the elder to study and compare their performance to that of others, through the implementation of the *social comparison* principle (Oinas-Kukkonen & Harjumaa 2009). In a similar vein, Oinas-Kukkonen and Harjumaa (2009) state through the *social facilitation* principle that if users can see others performing the target behavior, then they themselves are more likely to perform it.

As implied by the perception and ideas of the informants, a system incorporating this community of social learning around health would be highly beneficial to promoting healthy behavior and help to combat loneliness and a feeling of no social belonging in the elderly. Additionally, in any community or social group, there may be an incidence of peer pressure, which according to Oinas-Kukkonen and Harjumaa (2009) is another persuasive social factor (*normative influence*) that increases the likelihood of adopting a certain behavior.

Oinas-Kukkonen and Harjumaa (2009) also states *cooperation* and *competition* as strong persuasive influences. Competition is also stated through the informants in this study as a strong motivation for adopting health behavior or intensifying efforts toward a goal. A system that incorporates gamification and a reward system can leverage both of these principles as well as the *rewards* and/or *praise* dialogue design principles that as well have a persuasive power. If a system such as this incorporates the *recognition* social design principle by offering public recognition of rewards (or results of cooperation and competition) it may also act as a strong persuasive influence (Oinas-Kukkonen & Harjumaa (2009). Bravo et al. (2018) also states that gamification can encourage use of m-health systems.

Our findings show that the elderly may have a wish of being acknowledged for self-improving efforts, which further validates the inclusion of the recognition principle. Informants in the study also expressed that positive feedback had the effect of spurring them on towards reaching their self-improvement goals, while negative feedback was disliked and had the opposite effect. There was also a preference for encouragement toward a goal rather than discouragement of bad habits. As such, the praise design principle is highly encouraged, and care should be taken to not engage in scolding or punishments as it can be discouraging for reaching a set goal.

One could also argue that in a system incorporating public presentation of an individual's performance, negative results shouldn't be spread among the community. Rather, the focus should be on sharing positive results. This was also expressed by one informant. This could possibly promote a more positive and relaxed atmosphere in the community.

5.1 Elders' technology use

As implied by our findings and showed by prior research into technology use in the elderly (Dodd, Athauda and Adam 2017), there is a distinct lack of technology use in the elderly. This is possibly more prevalent in older elderly, where use of modern technology is close to non-existent. For this group, simply adhering to accessibility guidelines may not be enough as there may be a fundamental lack of understanding of the platform, or even resistance to adopting new technology. Negative attitudes to technology are seemingly present in the elderly. Informants in this study expressed ongoing difficulty with technology in some way or other that may reinforce a negative attitude, and one informant explicitly voiced strong resistance to technology and described it as a detriment to society.

For those elders that have at least some experience with using modern technology, their issues might be solved by adhering to good heuristics and accessibility guidelines. The findings seem to show that the younger bracket of elders are more familiar with technology. This technology experience gap could be rooted in a later introduction to technology; the younger bracket may have had more opportunities to adapt to the technology through its adoption through the workforce and other avenues.

There may be an argument for two different approaches for the elderly based on their prior knowledge of modern technology. Though, it would be impossible to generalize it to simply younger versus older, it is rather a question of one group needing a more significant onboarding process to effectively use the technology, while the other may be able to directly proceed to using it. There seem to be a general exclusion of the elderly group that have very low technological self-efficacy, as developed systems may take for granted that the user know and can use the technological platform and know the conventions used in it. Oftentimes, as was the case for the informants in this study, the elder sought help from relatives familiar with the technology. But having to consult others for basic use seems like a failure of the technology.

An onboarding process would effectively be an extension of the persuasion process. Adoption of new technology or platforms could be seen as a change in behavior, and thus if the persuadee is required to use an unfamiliar technology then this as well requires a persuasion process. We consider it ill-advised to proceed with attempts to change multiple behaviors at once, thus such a process should preferably be done sequentially. Care should be taken to not cause any initial friction during adoption of technology, as this may cause rejection. This also requires proper examination into

relevant usability principles and heuristics, to minimize occurrence of incidents and lower cognitive effort.

Because of the unfamiliarity with the patterns or conventions of the system in question for elderly with low technological self-efficacy (Dodd et al. 2017), designers should carefully consider where the meaning for certain symbols or interactions originated. Striving to keep symbols or interactions as close to real-world equivalents as possible as described by Inostroza et al. (2016) in SMASH2 is likely a good guideline to follow in this case.

Since the findings show a reliance of external help, the heuristic SMASH11 as described by Inostroza (2016) may be especially important to provide help and documentation directly where it is needed. This information should be obvious and easily accessed.

5.2 Research contribution

The research gap attempted to address in this study was the lack of understanding in the motivations and intentions of the elderly population in the use of persuasive m-health applications, as revealed by Orji and Moffat (2018). Previous research lacked a participatory approach to this issue (Orji & Moffat 2018), which was the primary approach to data collection used in this case study. Additionally, Seifert et al. (2017) expressed a need for investigation into m-health usability for older adults.

The findings of this study suggest that the primary motivating factor for healthy activity was social aspects, such as social pressure, competition and social belonging. Vice versa, demotivating factors included a lack of social support which by extension causes a decline in mental well-being that impacts health, as expressed by the informants. This implies that the most effective persuasive strategies are those that leverage social aspects for behavior and/or attitude change. In this context, the actual health goals are thus secondary to involvement in a social community.

The findings also imply that the main usability issues are derived from a lack of technological self-efficacy and lack of technology use. There already exist guidelines for making usable mobile interfaces for elderly, via general usability heuristics like SMASH (Inostroza et al. 2016) and usability issues and solutions specifically targeting the elderly (Dodd, Athauda and Adam 2017; Jin et al. 2007). The problems arise due to a fundamental lack of familiarity with the technology of the platform in question. This may imply that introducing new technology to a user should also be approached from a persuasive point of view, as trying to persuade a user to adopt new technology could be seen as persuasion to change behavior.

Our hope is that this contribution can aid the creation of persuasive m-health systems more adapted to the elderly population to improve their health and well-being.

5.3 Reflection of methodology

Below follows the personal reflections of the researchers in the execution and choice of methodology in the case study: semi-structured interviews, future workshop, revolutionary rapid prototyping, and usability evaluation.

5.3.1 Semi-structured interviews

Semi-structured interviews were chosen as it was deemed the most appropriate method for the context in order to satisfy the requirement of involving the target audience with participatory methods. An alternative to this could have been to base our knowledge of the target audience on prior research, though this would go against the participatory approach deemed necessary by Orji and Moffat (2018) to further the research on persuasive technology for the elderly.

Our impression of the semi-structured approach to interviews, with topics instead of set questions, is that it made for a more natural dialogue. The interview felt more casual and promoted a more relaxed atmosphere for the informant (and the interviewer). It also gives more room to the informant to speak freely and gave them a platform to air their concerns. It treats the informant with more dignity, like an equal, and emphasizes that their insight is valuable. Though, an interview guide was critical for this approach to function, to be able to cross off topics that have been exhausted and make sure the current topic is relevant.

We perceive that the environment of the interview (the informant's house) helped to promote a congenial atmosphere. Furthermore, the fact that the interviewer is invited to the informant's house may contribute to the informant feeling more in control over the interview given that the informant is the host and the interviewer is a guest. This arrangement presuppose a condition of mutual trust, but as mentioned, the option to choose location was given to the informant to assure they were comfortable with the environment. Considering all informants were at the least acquainted with the interviewer, the feeling of trust were likely high enough that choosing their own house as the location for the interview wasn't an issue.

Negatives perceived in the execution of the method was primarily because of low experience of doing interviews. This led to moments where the interviewer failed to inform the informant to refrain from talking in general or third-person, and instead focus on their own experiences. There were also some occurrences of closed and leading questions that encouraged specific answers (such as "Do you think X is Y?").

For the transcription, automated machine transcription was tested, but the technology failed to construct understandable sentences from the interview audio. Which led to simply doing by hand. Instead of reducing interview data to variables, the entire dialogue for the question and answer was kept to retain the context and original meaning. We deemed this was more beneficial than variables when analyzing low

amounts of qualitative data, as variables removes a lot of the context of the question. In addition, recording variable frequency was deemed not as useful when dealing with a low number of informants, as no proper conclusions can be drawn based on it.

5.3.2 Future workshop

Future workshop was chosen because of our prior experience in using the method. Its usefulness has been personally proved as a participator method well-suited to improving the situation of a specific target group. We contemplated using co-design as an alternative, but our experience is that this method is harder to recruit for and participants are often reluctant to whole-heartedly participate due to the perception that their sketching skills aren't good enough or that they don't have enough experience with the context or technology. This leads to a difficulty in collecting a good quantity of ideas.

Democratic discussion such as performed in a future workshop is, from personal experience, generally more efficient and provides a better result. We perceive that most people are not averse to discussion and letting their issues be heard or give ideas for solutions, in comparison to sketching solutions.

We found that leaving out technological classifiers from the workshop was beneficial because of the low technological self-efficacy in the participants. This was done as the interview data pointed at a low technological experience in the target audience. The location of the workshop (the home of one of the participants) contributed greatly to a more relaxed atmosphere.

Negative aspects of the workshop could be that sometimes the discussions went off-topic because of the fervor and emotional response to certain topics or issues raised. As the facilitator didn't want to be rude and directly interrupt, the unrelated topic was allowed to go unchecked until an appropriate moment arose to steer it back to the topic.

Additionally, the post-it notes were hard to read for the participants when put up on the wall. A possible improvement could be to use bigger post-it notes or put them on the table in front of all the participants. The workshop also went overtime by at least half hour; more care should be taken in approximating the stated time, or to make sure certain phases don't take too long. Further emphasizing and reminding the participants to provide a quantity of ideas might have been good to get more ideas.

5.3.3 Revolutionary rapid prototyping

Revolutionary rapid prototyping was chosen due to that it was a requirement to produce an interactive prototype in a relatively short time. No other alternatives were considered, though the exact process to use was discussed. The creation of a sitemap was a late addition to the process, as it was deemed difficult to progress from sketches

to wireframes without an overview of the system hierarchy and structure. We perceive the chosen process and its methods as effective for rapidly producing a relatively high-quality prototype. Though more efficient use of time and planning the production of a sitemap from the start would have improved its execution.

The storyboard sketches were also cut short because of time constraints. The original process included revisions of the sketches before progressing to wireframes, as described by Goodwin (2009). If given enough time, this would have been included as it would have made the wireframes less exploratory. Instead, the sketches acted more as loose guidelines for the wireframes, and the wireframing process was used almost like a second sketching phase. Though, Saffer's (2010) approach to annotate the wireframes with their intended behavior was deemed very useful to know the functionality, animations, guidelines and motivations behind certain elements prior to moving to making mockups and adding interactivity.

Due to that wireframes don't explore the three-dimensionality of shadows, color theory, or the effect that these have on the overall visual experience of the prototype, it led to layout changes in the mockups to better fit the visual hierarchy. The contrast checker was very useful for determining the correct contrast between text and background.

Finally, if given the time, it would have been interesting to explore more alternatives for solutions, or even multiple prototype for testing. This would make the usability evaluation, or the testing of the prototypes into a way to determine the design path to take rather than focusing on confirming usability.

5.3.4 Usability evaluation

Usability evaluation was chosen as an appropriate method as to confirm usability and receive feedback on the system from the target audience. Possible alternatives might include a heuristic walkthrough of the prototype (though this was used in the analysis step of the prototyping process), though that would go against the participatory approach for the case study.

The evaluation required a bit of a technical setup with linking to the prototype, which required a smartphone that was setup with the prototype. This resulted in the use of a test smartphone as to not require the participant to set it up on their smartphone.

It was also difficult for the participants to express themselves and put words on their feelings and thoughts in the think-aloud approach. This may be an issue of cognitive load. Meaning that it is difficult to perform the given task, analyze and interact with the prototype in addition to articulating their thought process. It may as well be an issue of technological self-efficacy. Given low experience in using smartphone applications and systems such as this, it may be difficult to verbalize their thoughts and interactions with the different concepts presented through the interface.

Though, we have no suggestions on how to improve the situation in this context; the think-aloud method may just be inappropriate in a context where cognitive load may be an issue and the participant has low experience in using the platform or technology of the prototype.

5.4 Reflection of validity and reliability

By recruiting participants via convenience sampling from one's own social network the data may have been affected through interpretations based on known or assumed characteristics of the participant. Though, it likely contributed to a more relaxed and possibly honest dialogue.

The European background of the facilitator and the project as a whole can also have created distance between the participant and its involvement. Also, since the facilitator had English as a second language, it may have affected the interpretation of data and statements by the participants as well as introduced further difficulty in dialogue.

6 Conclusion

This chapter concludes the thesis with a summary of the case study, the arguments supporting the answer to the research question, as well as suggestions for further research.

We have in this case study investigated the health behavior, motivations for healthy behavior, attitude to health, general goal setting behavior, needs, preferences, technological experience and self-efficacy, as well as usage of m-health systems of elderly individuals using qualitative and participatory methods. This was done to find effective persuasive strategies targeting elderly individuals to answer the research question:

How can persuasive technology promote usage of an m-health application by the elderly?

We believe to have identified the primary motivational factor to target with persuasive technology in order to promote usage of m-health applications. Findings suggested the primary motivational factor was social aspects. Lack of social involvement was shown to be a profound demotivating factor that had negative repercussions for health and well-being, both physical and mental. Using persuasive strategies that target social aspects may have a strong persuasive power to change the behavior and attitude of elderly individuals, as to promote healthy behavior. This could be done by creating elderly communities around activities for health as demonstrated by the prototype produced in this case study.

We believe it is crucial to possess a fundamental knowledge on how to fulfill the user's basic human needs like social belonging so that the m-health application doesn't merely provide utility. It should provide a complete experience to promote a good user experience and long-term use.

6.1 Future research

Because of the time span of the case study, this study has not been able to research the long-term effectiveness of persuasive strategies or measure the direct response to these due to the lack of ability to make a functional software prototype. It would be beneficial to examine the long-term effectiveness of the persuasive strategies recommended by this case study in changing the behavior and/or attitude to health of elders.

Additionally, no conclusive data could be gathered on the planning and goalsetting of the elderly as no such behavior was expressed by the participants in this study. Further research may be necessary on planning and goalsetting behavior in the elderly to adequately support them in creating reachable health goals.

It may be beneficial to perform a more detailed study in how persuasive technology can support the adoption of new or unfamiliar technology, especially for individuals with low technological self-efficacy.

The role of caregivers in m-health systems were not explored in this study, but the involvement of them in a system like this is highly recommended according to Bravo et al. (2018). Certainly, research on m-health systems that involves caregivers exist as demonstrated by Bravo et al. (2018), but in the case of an m-health system utilizing persuasive technology that focuses on social aspects and forming an elderly community to promote healthy behavior, this topic is less explored. Further research on the role of caregivers in this context is warranted to explore the usefulness of intertwining this type of m-health system closer with the healthcare system.

References

Arvola, M. (2014). Interaktionsdesign och UX: om att skapa en god användarupplevelse. Lund: Studentlitteratur AB.

Asmussen, R. (2019). *Man Taking Selfie* [photo]. https://www.pexels.com/photo/adult-aged-boy-facial-expression-334029/ [2019-05-22]

Bravo, J., Hervás, R., Fontecha, J. & González (2018). m-Health: Lessons Learned by m-Experiences. In: *Sensors*, 18(5). DOI: 10.3390/s18051569

CELTIC-NEXT (2016). *FROm empoweriNg To Viable Living.* https://bscw.celticplus.eu/pub/bscw.cgi/d16903/FRONT-VL_start_lq.pdf

Creative Commons (2019). *Attribution 4.0 International (CC BY 4.0).* https://creativecommons.org/licenses/by/4.0/ [2019-06-02]

Creswell, J. W., & Creswell, J. D. (2018). *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*. 5th ed. California: Sage Publications.

Dodd, C., Athauda, R. & Adam, M. T. P. (2017). Designing User Interfaces for the Elderly: A Systematic Literature Review. In: *Australasian Conference on Information Systems.* Hobart, Australia.

Edwards, R. & Holland, J. (2013). *What is qualitative interviewing?* London: Bloomsbury.

Font Awesome (2019). *Icons*. https://fontawesome.com/icons?d=gallery&m=free [2019-05-25]

Goodwin, K. (2009). *Designing for the Digital Age: How to Create Human-Centered Products and Services.* Indianapolis: Wiley.

Hassenzahl, M. (2008). User Experience (UX): Towards an experiential perspective on product quality. In: *IHM '08 Proceedings of the 20th Conference on l'Interaction Homme-Machine*. Metz, France 2-5 september 2008, pp. 11-15. DOI: 10.1145/1512714.1512717

Inostroza, R., Rusu, C., Roncagliolo, S., Rusu, V. & Collazos, C. A. (2016). Developing SMASH: A set of SMArtphone's uSability Heuristics. In: *Computer Standards & Interfaces*, 43, pp- 40-52. DOI: 10.1016/j.csi.2015.08.007

International Organization for Standardization (ISO) (2018). *ISO 9241-11:2018 Ergonomics of human-system interaction: Part 11: Usability: Definitions and concepts*. 2nd ed. Switzerland: ISO.

Jin, Z. X., Plocher, T. & Kiff, L. (2007). Touch Screen User Interfaces for Older Adults: Button Size and Spacing. In: *4th International Conference on Universal Access in Human-Computer Interaction, UAHCI 2007*. Beijing, China, pp. 933-941. DOI: 10.1007/978-3-540-73279-2_104

Mozilla (2019*). mozilla/fxemoji: FxEmojis – a friendly emoji set from Mozilla*. https://github.com/mozilla/fxemoji [2019-06-02]

Norman, D. (2013). *The Design of Everyday Things.* 2nd ed. New York: Basic Books.

Oinas-Kukkonen, H. & Harjumaa, M. (2009). Persuasive Systems Design: Key Issues, Process Model, and System Features. In: *Communications of the Association for Information Systems*, 24(1), pp. 485-500. DOI: 10.17705/1CAIS.02428

Orji, R. & Moffatt, K. (2018). Persuasive technology for health and wellness: State-of-the-art and emerging trends. In: *Health Informatics Journal*, 24(1), pp. 66-91. DOI: 10.1177/1460458216650979

Oliveira, R., Cherubini, M. & Oliver, N. (2010). MoviPill: improving medication compliance for elders using a mobile persuasive social game. In: *UbiComp '10 Proceedings of the 12th ACM international conference on Ubiquitous computing*. Copenhagen, Denmark, pp. 251-260. DOI: 10.1145/1864349.1864371

Overmyer, S. P. (2002). Revolutionary vs. Evolutionary Rapid Prototyping: Balancing Software Productivity and HCI Design Concerns. In: *Proceedings of the 4th International Conference on Human-Computer Interaction.* Stuttgart, Germany, pp. 303-308.

Pixabay (2019). *Woman Smiling* [photo]. https://www.pexels.com/photo/adult-elder-elderly-enjoyment-432722/ [2019-05-22]

Region Jönköpings län (2017). *Passion för livet: Handledarmaterial*. Jönköping: Region Jönköpings län. https://folkhalsaochsjukvard.rjl.se/api/Evolution/pdf/3097e3b0-e2a0-49d4-aaf8-594970e2af4f

Rosenfeld, L., Morville, P. & Arango, J. (2015). *Information Architecture: For the Web and Beyond*. Sebastopol: O'Reilly Media.

Saffer, D. (2010). *Designing for Interaction: Creating Innovative Applications and Devices.* 2nd ed. Berkeley: New Riders Publishing.

Salim, M. J. M., Ali, N. M. & Noah, S. A. M (2017). Mobile Application on Healthy Diet for Elderly based on Persuasive Design. In: *International Journal on Advanced Science, Engineering and Information Technology*, 7(1), pp. 222-227. DOI: 10.18517/ijaseit.7.1.1725

Seifert, A., Schlomann, A., Rietz, C. & Schelling, H. R. (2017). The use of mobile devices for physical activity tracking in older adults' everyday life. In: *Digital Health*, 3. DOI: 10.1177/2055207617740088

Sveriges Kommuner och Landsting (2018). *Beteenden och behov hos personer i kontakt med vården: diskussionsunderlag för utveckling och innovation.* Stockholm: Sveriges Kommuner och Landsting. https://skl.se/download/18.5e588ed415aa6ecabde9f48f/1489484509965/Beteenden%20och%20behov%20hos%20personer%20i%20kontakt%20med%20v%C3%A5rden.pdf

Tague, N. R. (2005). *The Quality Toolbox.* 2nd ed. Milwaukee: ASQ Quality Press.

Taylor, M. J., McNicholas, C., Nicolay, C., Darzi, A., Bell, D. & Reed, J. E. (2013). Systematic review of the application of the plan–do–study–act method to improve quality in healthcare. In: *BMJ Quality & Safety*, 23(4), pp. 290-298. DOI: 10.1136/bmjqs-2013-001862

Vidal, R. V. V. (2005). *The Future Workshop: Democratic Problem Solving.* Lyngby: Technical University of Denmark.

World Health Organization (WHO) (2015). *World report on ageing and health.* Geneva: World Health Organization. https://apps.who.int/iris/handle/10665/186463

World Wide Web Consortium (W3C) (2018). *Web Content Accessibility Guidelines (WCAG) 2.1*. https://www.w3.org/TR/WCAG/#contrast-enhanced [2019-05-28]

Appendices

Requirements specification

The following requirements are based on an introductory letter provided by Sigma Technology and Linnaeus University, as well as requirements inquiries during project meetings with these parties. The requirements have been further specified and clarified to aid measurability. The requirements have been demarcated to requirements for the project's methodology and purpose as shown in Table A.1 and functional requirements for the prototype as shown in Table A.2.

Table A.1 – Project requirements

	Requirement	***Description***
P1	Include SKL behavioral groups	The project shall take into account the four behavioral groups of people in contact with the healthcare system as put forth by Sveriges Kommuner och Landsting (SKL) when developing solutions.
P2	Include PDSA cycle	The project shall include the plan-do-study-act (PDSA) cycle in the literature review and theoretical framework.
P3	Elderly target audience	The target audience for informants shall be elderly (65+) or otherwise frail individuals that live at home.
P4	Participatory approach	The project shall directly involve the target audience in the design process via participatory methods.
P5	Produce a mobile application GUI	The project shall produce a graphical user interface (GUI) for a mobile application that demonstrates (visually and through interactions) the required functionalities (see Table A.2) and satisfies the needs of the target audience.

Table A.2 – Functional requirements

	Requirement	***Description***
F1	Login system	The application shall include a login system using a username and password login method.
F2	Data visualization	The application shall be able to visualize data using diagrams.
F3	Time series data	The application shall be able to organize, and display data based on time.
F4	Goal setting	The application shall enable the user to set (multiple) goals and the application shall be able to detect process toward the goals. Accomplished goals shall be archived.
F5	PDSA cycle plan	The application shall support goal setting, progress and planning with the PDSA cycle. The application shall enable the user to edit their PDSA plan and view help documentation about PDSA.
F6	Self-tests	The application shall enable the user to perform (multiple different) self-tests. Self-test data is integrated into the user's dataset and shall be able to be displayed individually and as part of a data summary.

Personas

Alice

Figure B.1 – Alice persona. Free from Pixabay (2019).

- 89 years old
- Tech-resistant; has smartphone, but only uses for calling; need outside help for using new technology
- Lives alone in a house
- Doesn't enjoy gamification or competition; prefers cooperation and face-to-face interactions
- Lacking social network; suffer from loneliness; no children; good contact with siblings
- Has a one-day-at-a-time approach to goalsetting; spontaneous
- Frail; osteoporosis; fall risk
- Traditional health behavior: manage by herself, rarely complains or consult doctors for health problems, unworried; steady care contact because of chronic condition

Function needs:

- Rigorous guides, demonstrations and help to use the interface and functions.
- Community and social engagement to motivate activity and combat loneliness
- Motivation to being more involved in personal health; awareness of health status
- Goalsetting help (goalsetting guide to determining a fair goal); daily activity schedule or overview

Hubert

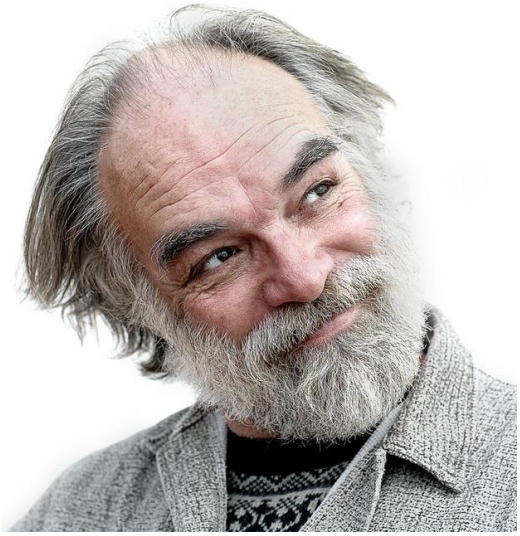


Figure B.2 – Hubert persona. Free from Asmussen (2019).

- 67 years old
- Moderate tech interest; has smartphone and uses some apps; can mostly manage new technology by himself
- Lives with wife in a house
- Enjoys gamification and competition
- Strong social network; has kids
- Sets structured goals and plans
- Moderately healthy; need proactive activity
- Health conscious; worsening health causes health anxiety; seeks health advice and solutions independently; no steady care contact

Function needs:

- Detailed information and visualization of activity statistics.
- Detailed goalsetting and planning functions for activity motivation; improvement tracking
- Overview and control over health to alleviate anxieties and to feel in-control
- Promote positive attitude to health over anxiety-inducing
- Motivation via gamification and competition

Context scenarios

Context scenario for Alice

Alice starts the day with exercises provided by her doctor last week. Uninspired by the lack of belief in the benefits and reason of performing these exercises she starts doing prescribed exercises with a rubber band. But after just a few repetitions she lets go of the rubber band and moves into the living room and sits down in front of the television and watching the morning news instead.

Alice's two siblings passed away over ten years ago and her old friend and neighbor, Susan, moved into an assistant living complex a few months ago. Earlier in life Alice had a fulfilling social life but her age has come with an increased sensation of loneliness. Alice don't have any kids but been having a close relationship with her nephew over the years and always celebrated holidays and had family dinners regularly. Ever since Alice's sister passed away Alice been taking the role as a grandmother. But the distance to her relatives is over an hour and, since Alice no longer is able to drive a car, the visits have been more spaced out than what she would like.

Alice's smartphone dings and the vivid text says; "Susan walked 2000 steps". Alice opens the notification and sees that Susan has started the day with a morning walk and is ahead of both her and her nephew, as well as his two children who are in the competition. There is no actual reward but it usually on of the main topics during their Sunday dinners that occurs once or twice a month and act as sufficient motivation for Alice to strap on her walking shoes and take a brisk walk around the block. She does not want to be the one that is last by taking the fewest steps at the end of the week. After finishing her walk around the block, a sense of self accomplishment makes her do a couple of additional exercises with the rubber band before taking a well-deserved nap. While in bed she that both her nephews' children have commented on the morning walk; "Good job Grammie! <3" and "I better get going! Love you!". Alice falls asleep with a smile on her face.

In the afternoon, the feeling of loneliness is present. Alice opens her phone and, as instructed by her nephews' kids, scrolls through the social events in the vicinity. And finds there is a bocce-ball tournament arranged by the elderly couple down the street Alice announce her willingness to partake and sees that there is another person that also will partake. Alice is thrilled to have this opportunity, puts on her fancy jacket that she rarely wears and struts down the street to meet her new friends.

Context scenario for Hubert

Recently, Hubert have been becoming increasingly aware of his own health and mortality because of the state of his health changing due to aging. Hubert considers himself naturally health conscious and active in his own well-being. But he finds himself stuck in a negative spiral of health anxiety that has been making him excessively morose and hyper-aware of every ache and feeling in his body.

His attempts at breaking this spiral has been unsuccessful and doctor visits have been unhelpful; merely stating it is a natural part of aging. Hubert has tried to combat deteriorating health by exercises, but he finds it hard to continuously motivate himself to keep to a daily exercise schedule and his attempts at sticking to his plan falls apart multiple times, which just makes him frustrated with himself.

While chatting with his friend George, he brought up his frustrations about his current situation, and his friend recommended that he installed the FRONT-VL application. As a current user of the application, his friend explained it's quite useful for structuring health improvement, and even has somewhat of a local community around that through the app. Hubert acted interested but dismissed it and changed the topic.

Later that day, Hubert recalled the conversation and went ahead to check out the application. He made an account and was prompted to set up an activity goal. He chose to set 5000 steps per day as a goal. The application prompted him to set his current daily steps, due to the absence of data, and the application then presented a suggested plan to reach this goal over time, including spaced out sub-goals. If he could keep 5000 steps per day over a week at the end, the goal would be completed, and he would get this goal as achievement ribbon to be displayed in his profile. Hubert continued going about his day.

The next morning, he got a reminder for the number of steps he needed to reach that day through his goal plan. He didn't think much about it, and rapidly forgot about it after his morning coffee and just went about his day as normal. At the end of doing yardwork with his wife in the afternoon, he got a notification on his smartphone. The FRONT-VL application congratulated him on reaching half-way to his daily goal. His wife perked up and wondered what it was, and he explained to her. She expressed interest, and he helped set up a plan for her as well. Together they took a late afternoon walk to complete his daily goal.

The daily reminders and progress reports spurred him on to eventually reach his goal. Now he just needed to keep it to get his ribbon. He decided to utilize the community challenge function to motivate him through the last hurdle. Over the week, he challenged both his wife and George for the highest step counter. He managed to get 8000 steps against George, but still lost. In George's profile he could see his daily average was over 7000, so he guessed he shouldn't be too surprised. The application

congratulated him for the highest number of steps in one day the day after. Eventually, Hubert completed his goal and received his ribbon to display publicly. He could see a visualized summary of his performance trying to reach the goal. He had some slip-ups where he was below the daily goal, which were noted in the summary, as well as the ones well-above the average.

He realized he hadn’t been that down about his health these last weeks, and the application had helped him maintain a more positive outlook on health and activity, as well as giving more structure and support for keeping to a goal. He felt he was more in control of his health. He went ahead and set up a new goal; maybe he can reach 7000 steps like George?

Key path scenarios

Table D.1 – Alice key path scenarios

Context scenario excerpts	***Functional needs***	***Functional elements***
Alice's smartphone dings and the vivid text says; "Susan walked 2000 steps". Alice opens the notification and sees that Susan has started the day with a morning walk and is ahead of both her and her nephew, as well as his two children who are in the competition.	- System should know the requirements of the user. - System should be able to read sensors for step counting and store this data - System should be able to keep track of other participants - System should be able to calculate data from sensors - Notify the users of the system	- Push notifications in an appropriate format for the user. - Display data of the participants involvement - Personal settings (Settings page/view)
While in bed she that both her nephews' children have commented on the morning walk; "Good job Grammie! <3" and "I better get going! Love you!".	- System should make it possible for the user to comment on performances.	- Display comments - Create comment - Notify the user when - See other users' performances
Alice opens her phone and, as instructed by her nephews' kids, scrolls through the social events in the vicinity. And finds there is a bocce-ball tournament arranged by the elderly couple down the street Alice announce her willingness to partake and sees that there is another person that also will partake.	- See what events that are available based on geographic location. - Announce interest in partaking in an event. - Filter events - Create events	- Events view with a search and filter options - Event information; time, place, creator, general information and other participants. - Event creator

Table D.2 – Hubert key path scenarios

Context scenario excerpts	***Functional needs***	***Functional elements***
He made an account and was prompted to set up an activity goal. He chose to set 5000 steps per day as a goal. The application prompted him to set his current daily steps, due to the absence of data, and the application then presented a suggested plan to reach this goal over time, including spaced out sub-goals. If he could keep 5000 steps per day over a week at the end, the goal would be completed, and he would get this goal as achievement ribbon to be displayed in his profile.	- Login/account creation system - System should be able to automatically create goal plans from input and user data - Set/customize plans, goals and sub-goals. - Achievement system	- Account creation page(s) - First-time user goalsetting prompt - Goal-type input list - Goal completion condition input - Goal and plan visualization - Sub-goal input - Achievement ribbon preview
The next morning, he got a reminder for the number of steps he needed to reach that day through his goal plan. He didn't think much about it, and rapidly forgot about it after his morning coffee and just went about his day as normal. At the end of doing yardwork with his wife in the afternoon, he got a notification on his smartphone. The FRONT-VL application congratulated him on reaching half-way to his daily goal.	- Daily goal notifications - Progress report and encouragement	
He decided to utilize the community challenge function to motivate him through the last hurdle. Over the week, he challenged both his wife and George for the highest step counter. He managed to get 8000 steps against George, but still lost. In George's profile he could see his daily average was over 7000, so he guessed he shouldn't be too surprised.	- Online community - receive and send challenges to users	- Community hub - Application member list - Challenge interface (send/accept) - Challenge results screen - User profile screen - Public daily step info
The application congratulated him for the highest number of steps in one day the day after. Eventually, Hubert completed his goal and received his ribbon to display publicly. He could see a visualized summary of his performance trying to reach the goal. He had some slip-ups where he was below the daily goal, which were noted in the summary, as well as the ones well-above the average.	- Personal highscore system - Automatic plan/goal summary and visualization	- Personal highscore record/list - Set public display ribbons on profile - Ribbon information screen - Completed goal assessment and overview screen

Storyboard sketches

Alice

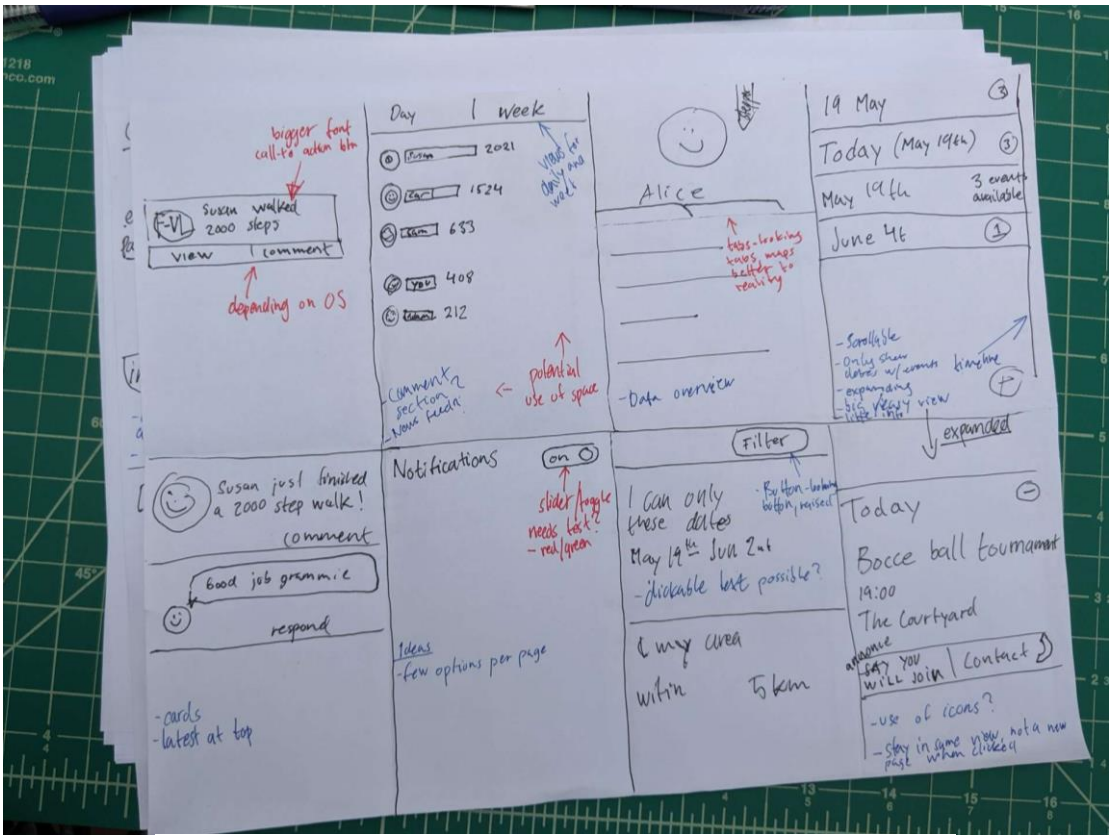


Figure E.1 – Alice storyboard sketch #1

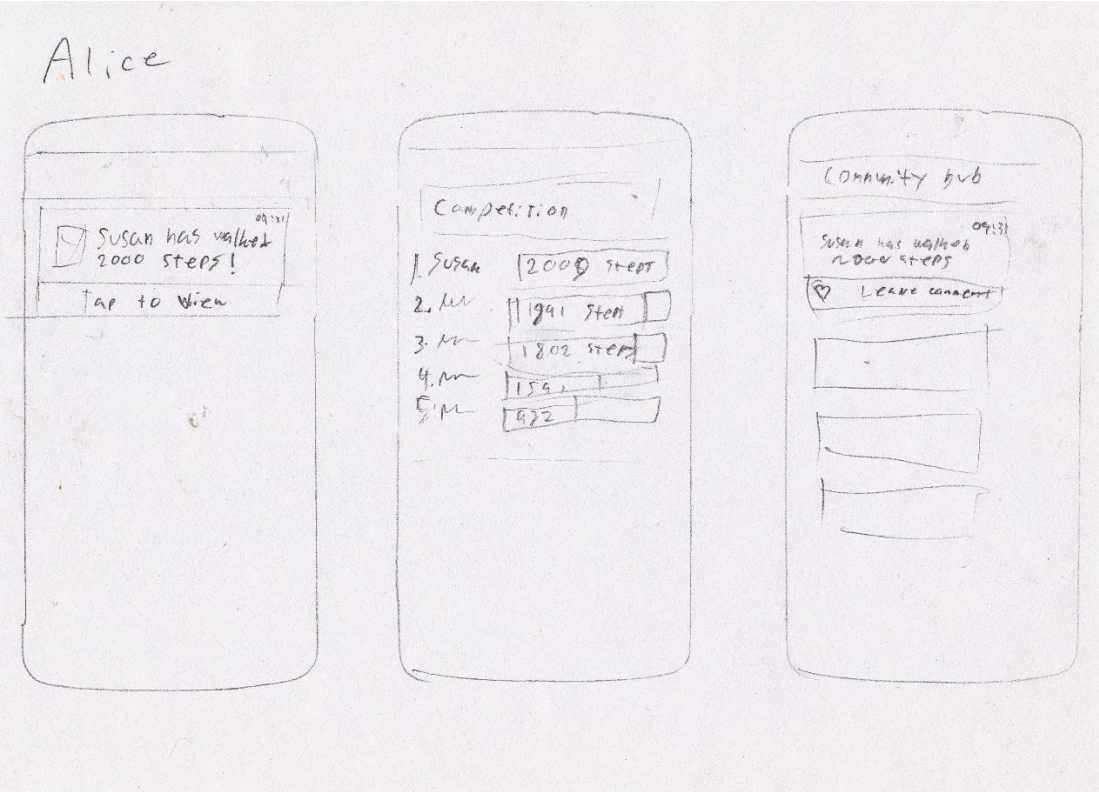


Figure E.2 – Alice storyboard sketch #2. Notification from friend (left). Active competition (middle). Community hub post/comment flow (right).

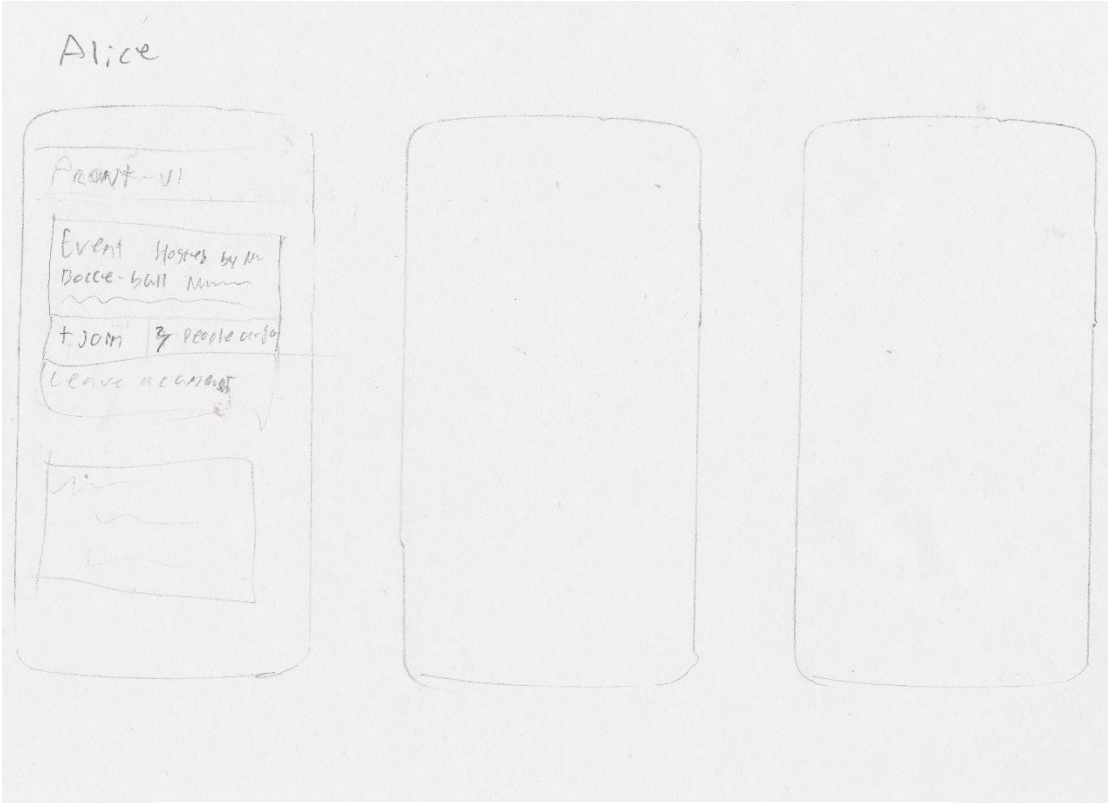


Figure E.3 – Alice storyboard sketch #3. Community hub event post (left).

Hubert

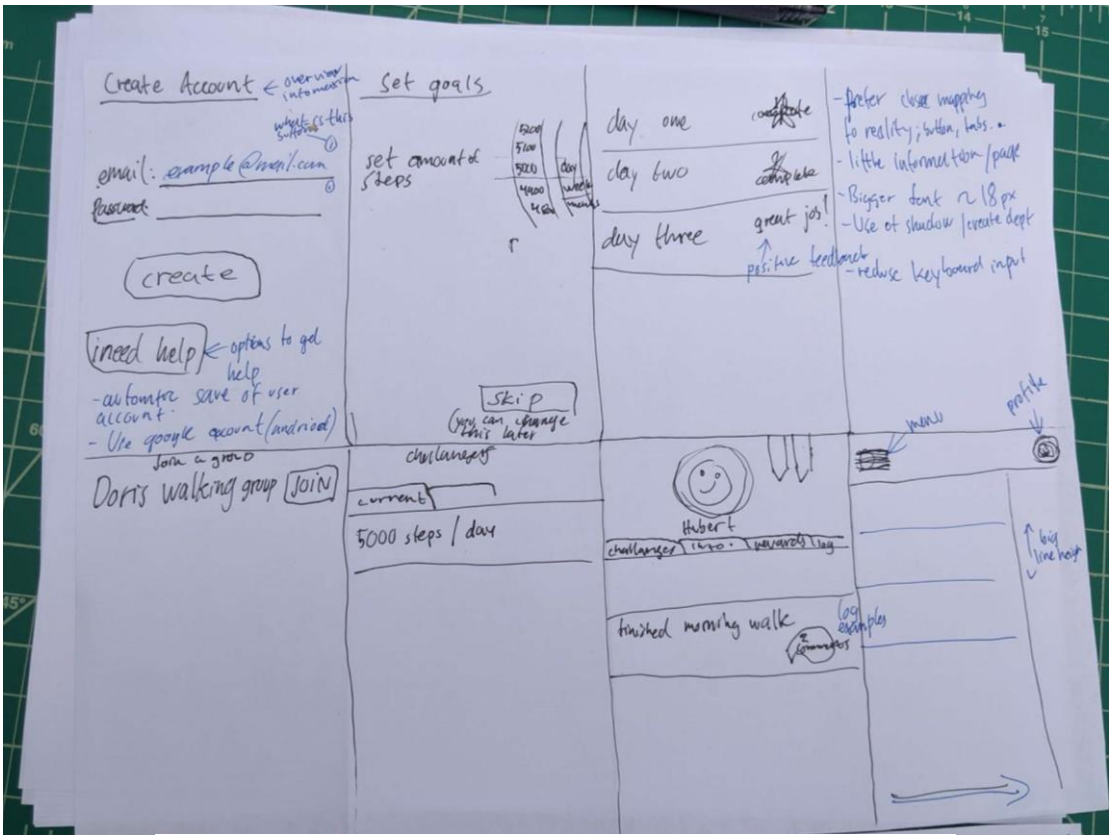


Figure E.4 – Hubert storyboard sketch #1

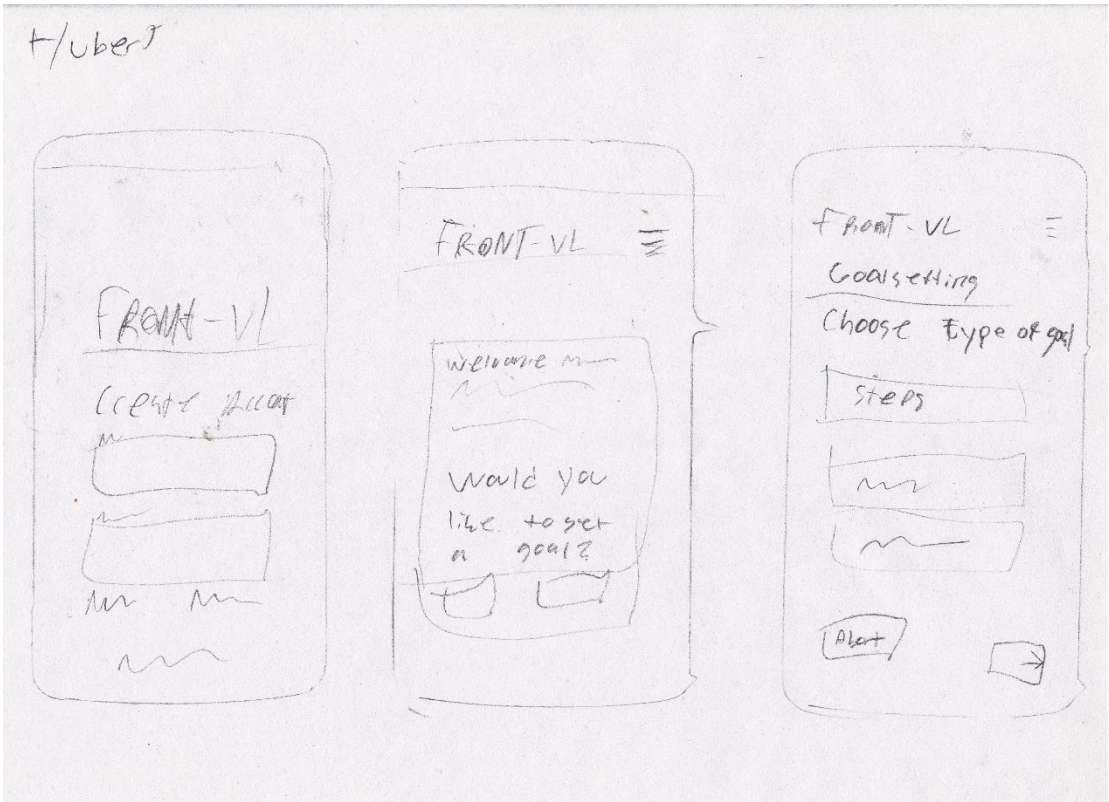


Figure E.5 – Hubert storyboard sketch #2. Log in screen (left). Welcome message on account creation with shortcut to setting goal (middle). Goalsetting wizard goal type selection (right).

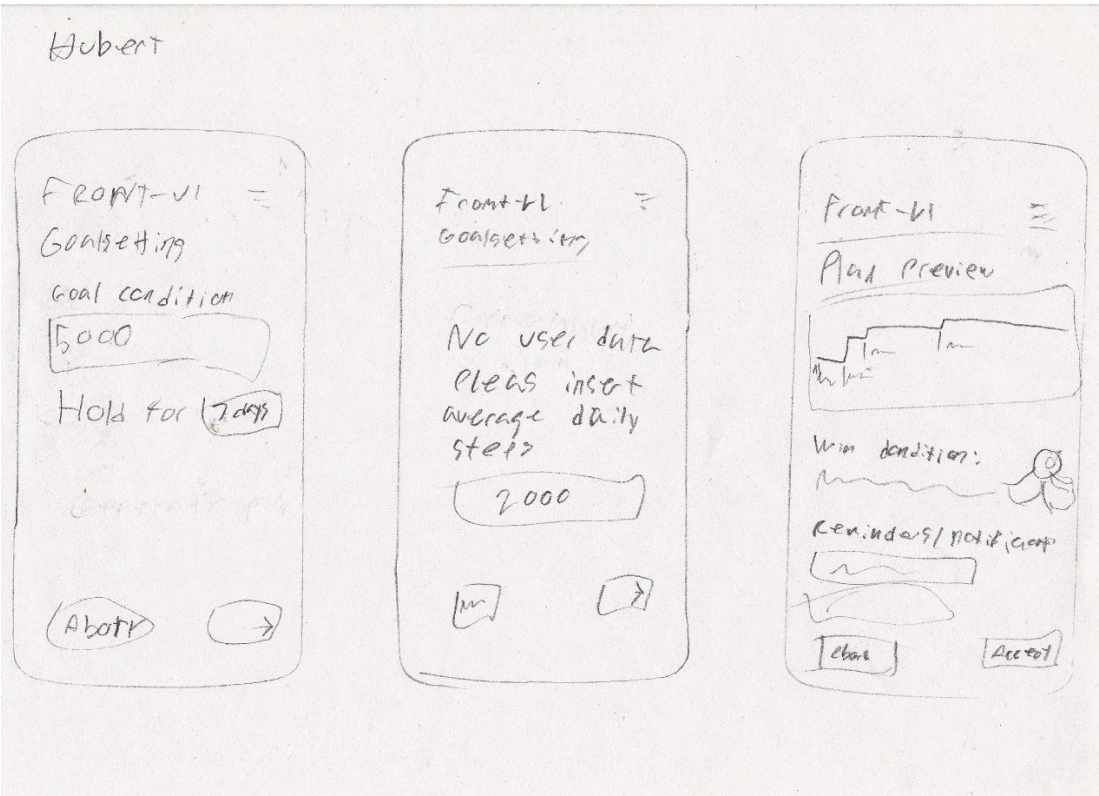


Figure E.6 – Hubert storyboard sketch #3. Goalsetting wizard goal win condition (left). Average step data input when no user data found (middle). Edit interface for computer generated (PDSA) goal plan (right).

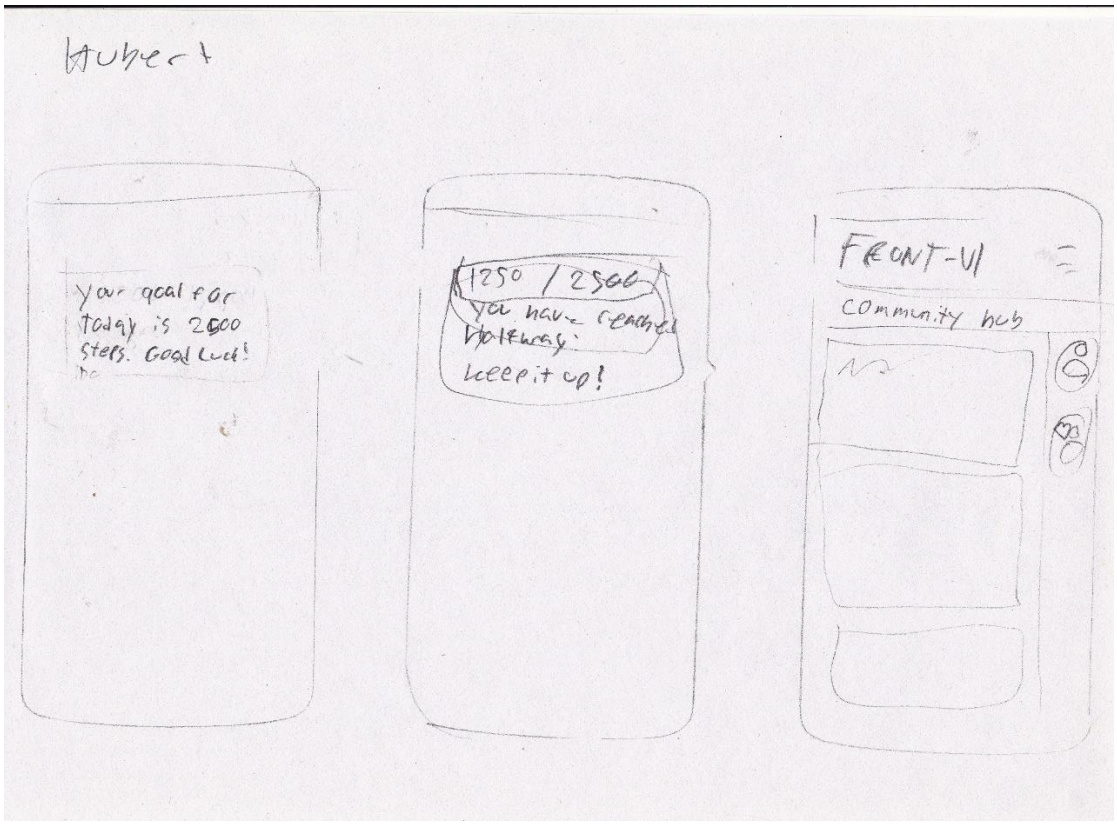


Figure E.7 – Hubert storyboard sketch #4. Daily goal notification (left). Half-way daily progress praise (middle). Community hub post/comment flow (right).

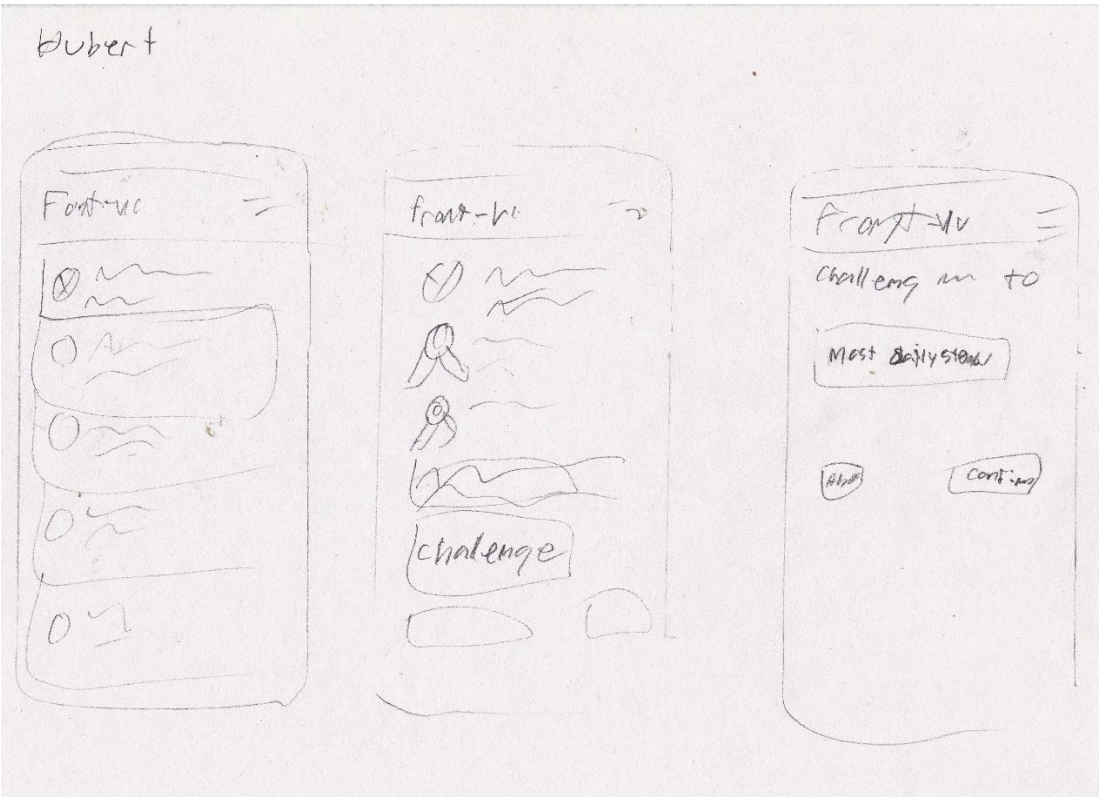


Figure E.8 – Hubert storyboard sketch #5. User list (left). User profile (middle). Challenge wizard challenge type (right).

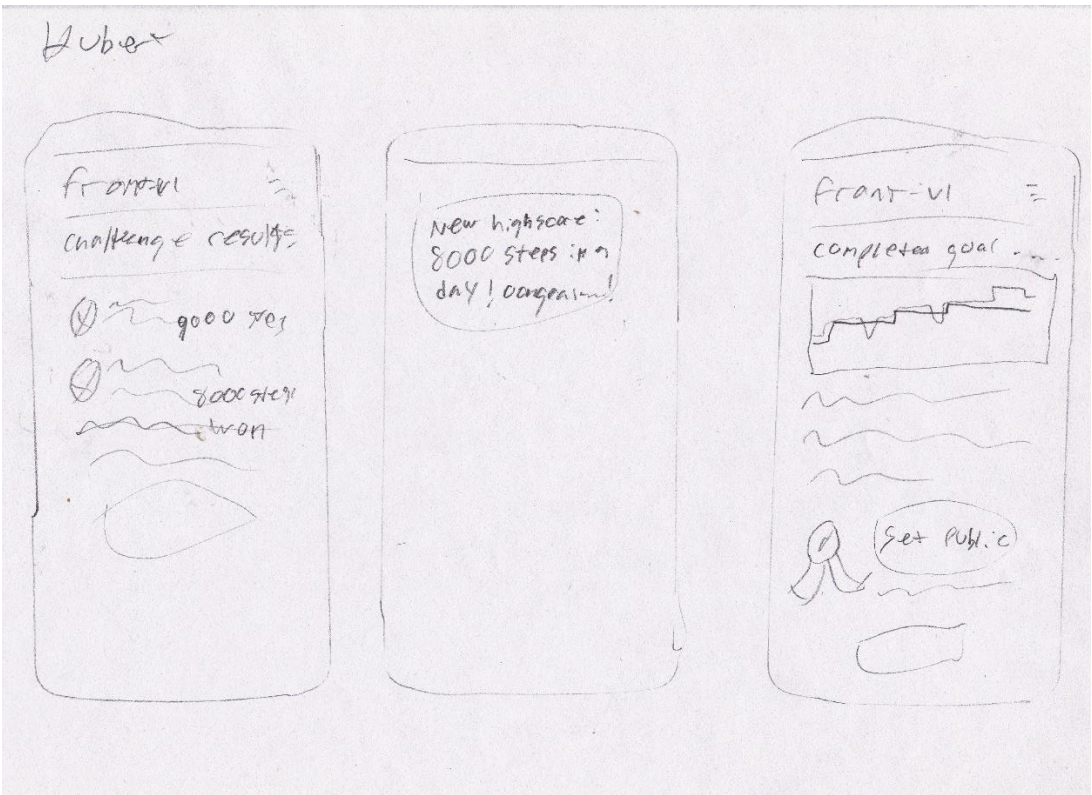


Figure E.9 – Hubert storyboard sketch #6. Challenge results (left). New personal highscore notification (middle). Completed goal evaluation and overview (right).

Sitemap

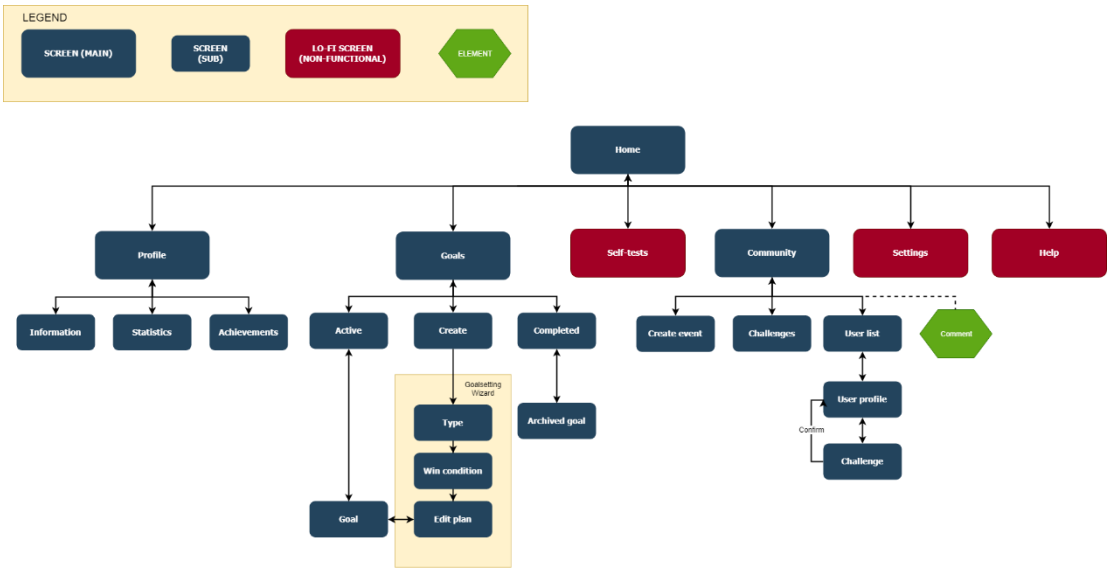


Figure F.1 – Sitemap

Wireframes

The figures below is all of the wireframe screens produced through the revolutionary rapid prototyping.

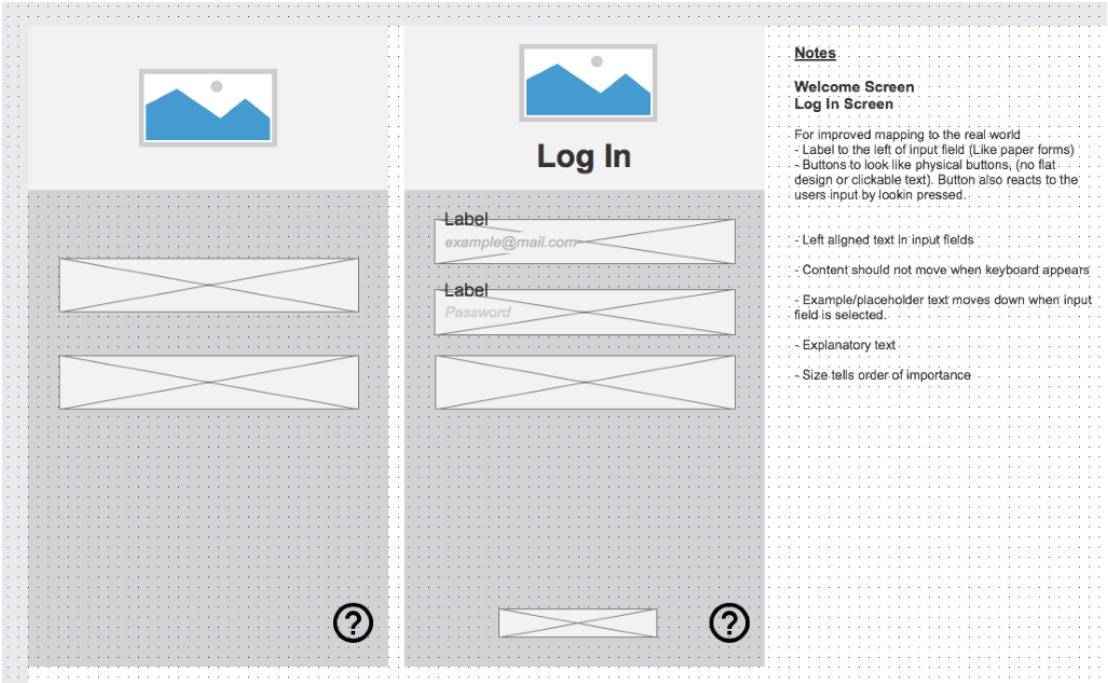


Figure G.1 – Welcome screen (left) and log in screen (right)

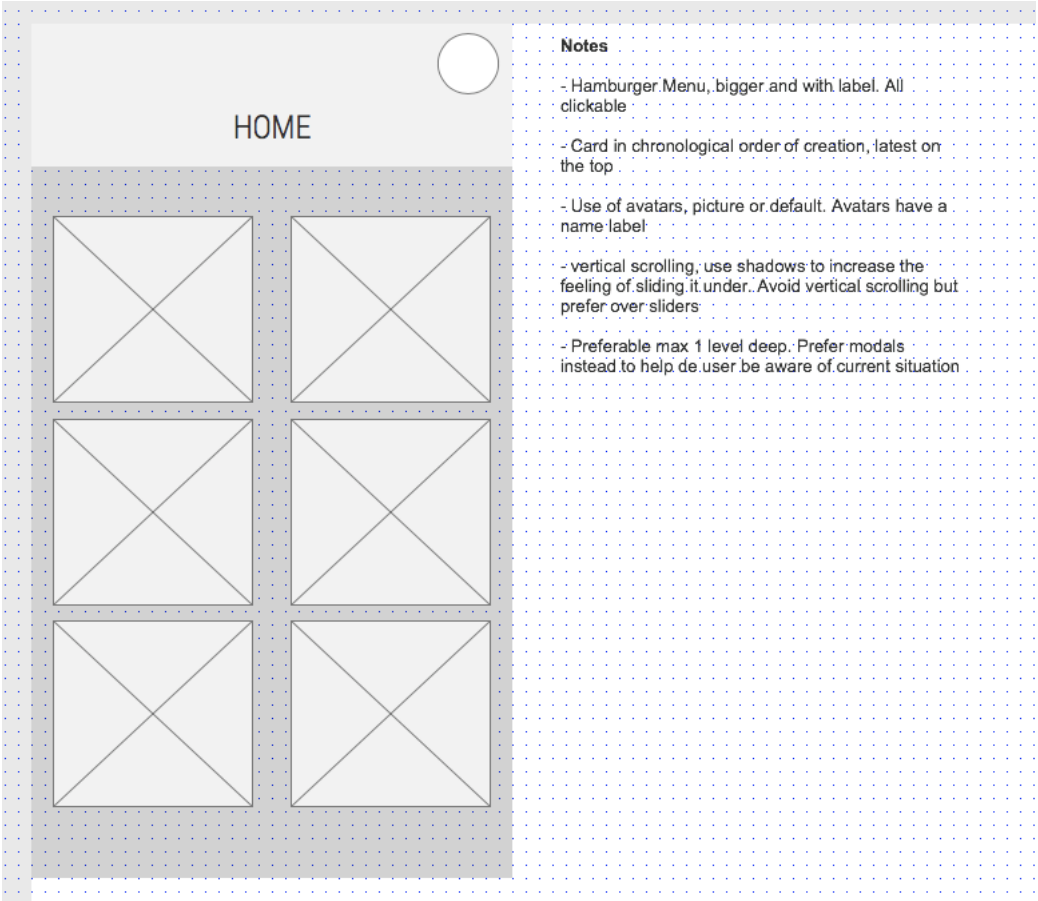
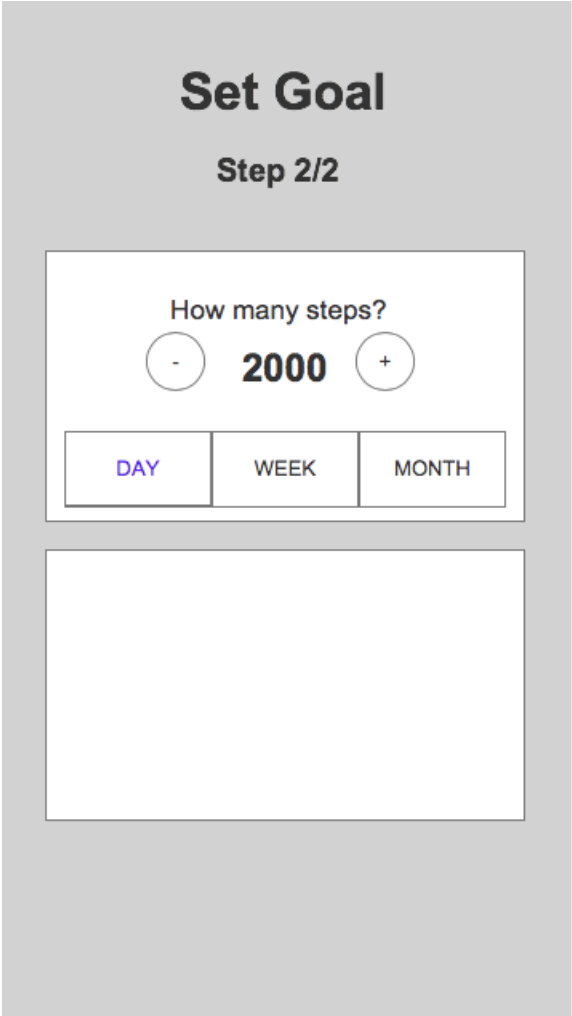


Figure G.2 – Home screen



Notes

- Steps Modal. Scroll wheel/spinner 100 steps increments
- Long presses should be slowly increasing
- Gives feedback and confirming updated/provided number

Figure G.3 – Goalsetting screen for steps

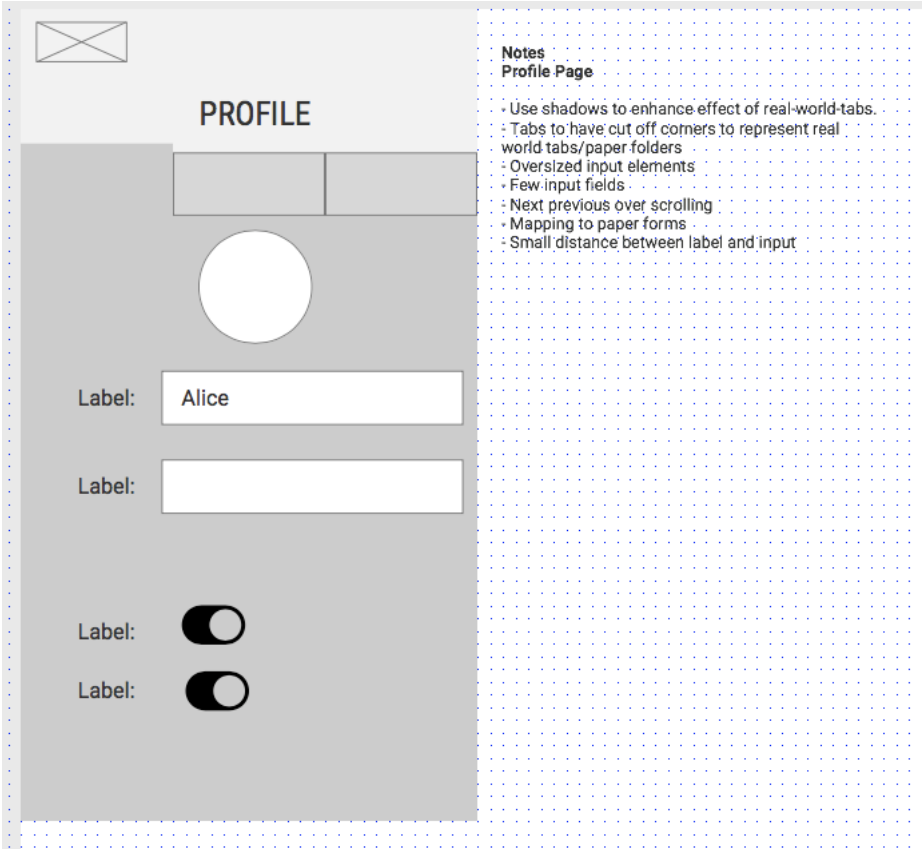


Figure G.4 – Profile screen

Figure G.5 – Profile modal screen

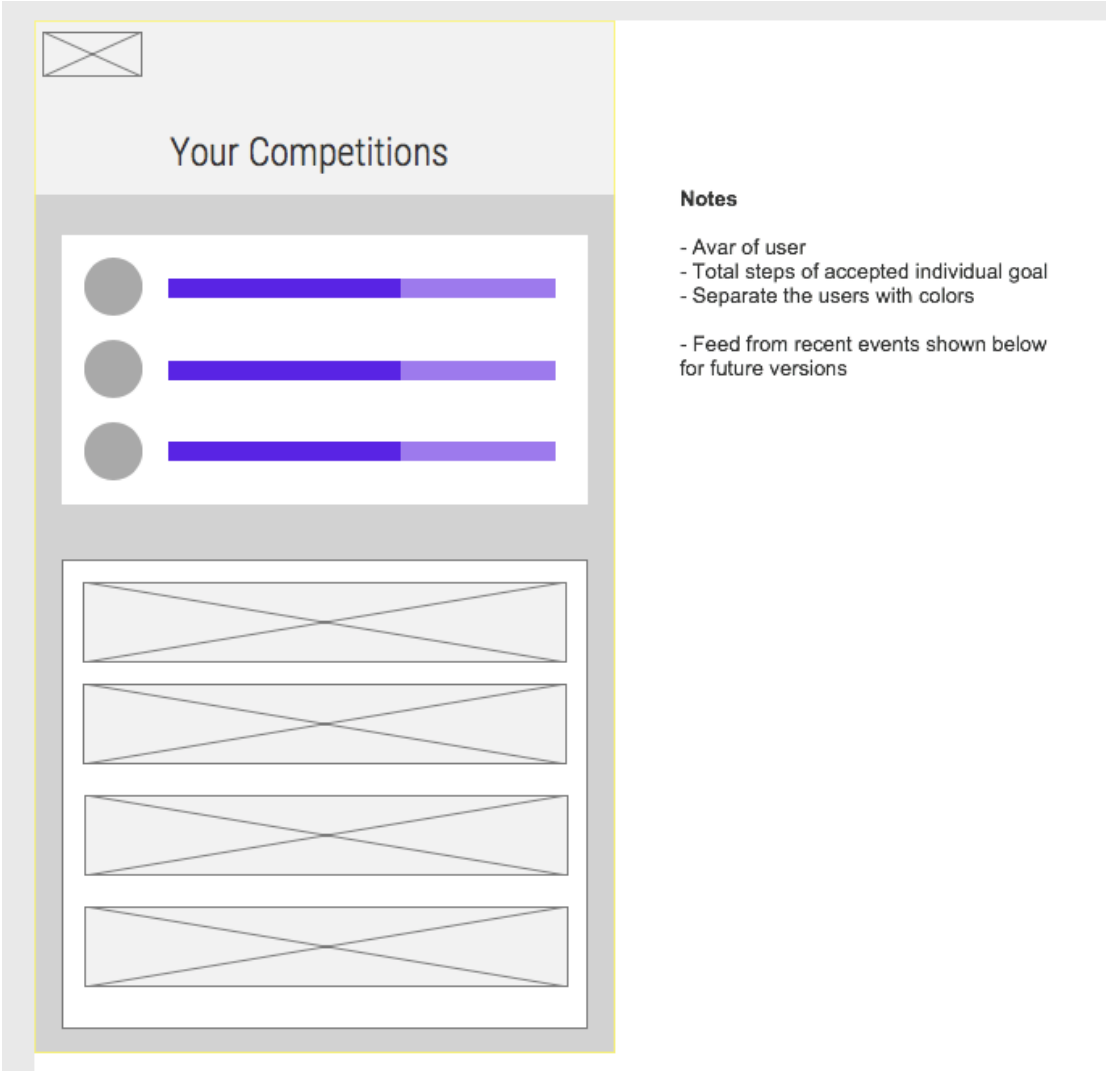


Figure G.6 – Competition screen

Rapid prototype

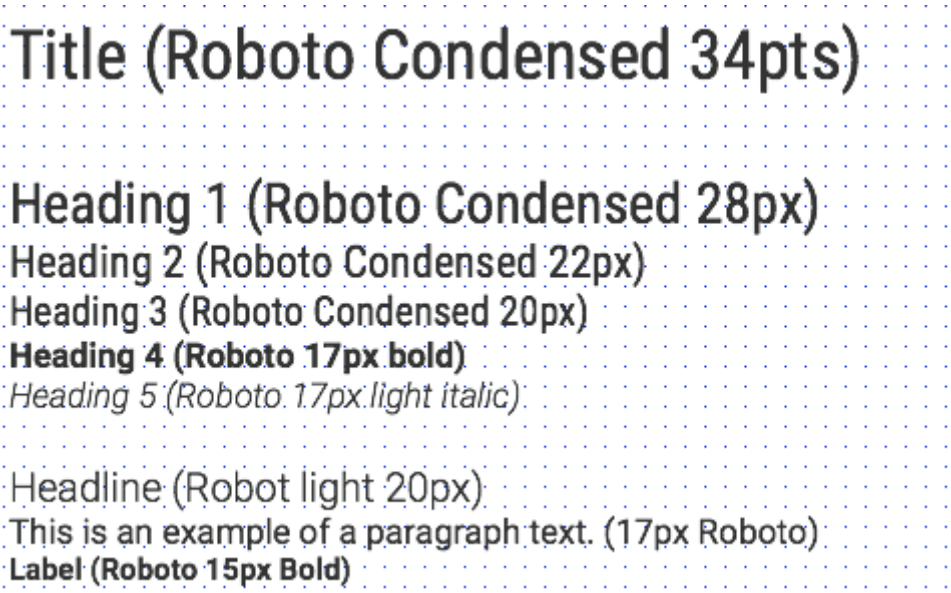


Figure H.1 – Font design specification

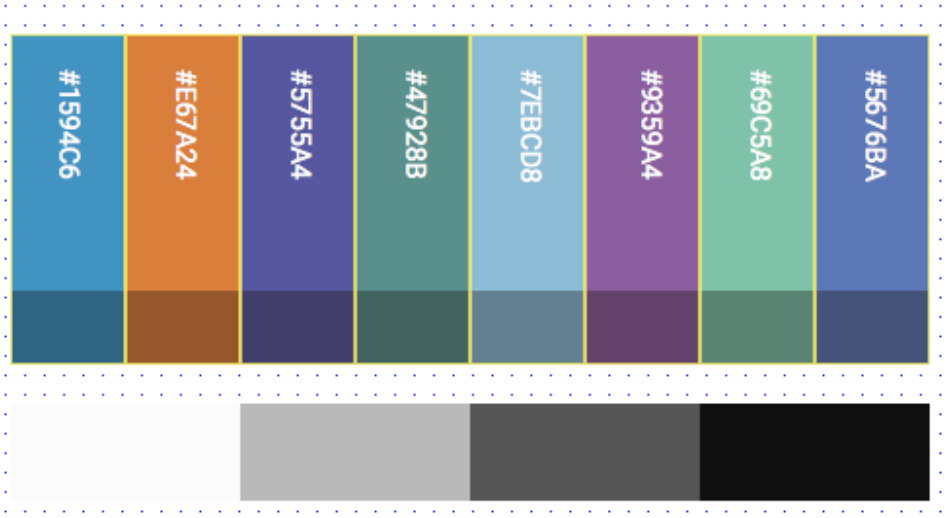


Figure H.2 – Color design specification

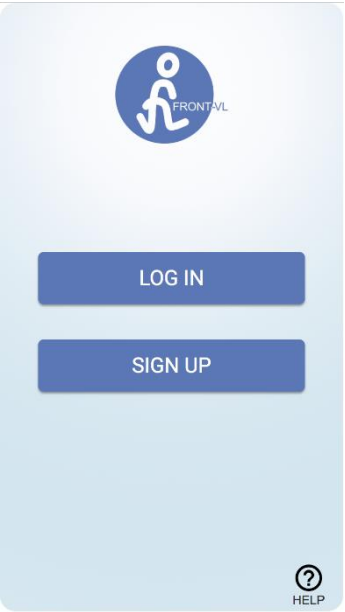


Figure H.3 – Welcome screen

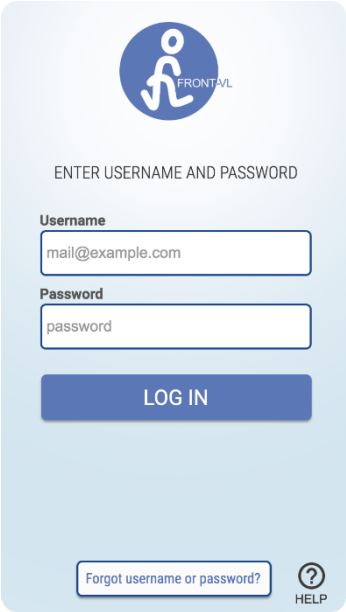


Figure H.4 – Log in screen

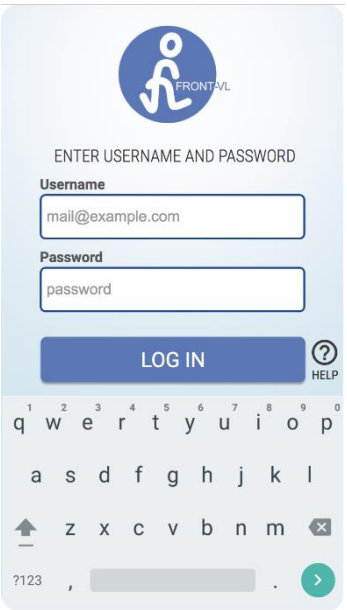


Figure H.5 – Log in screen (with keyboard)

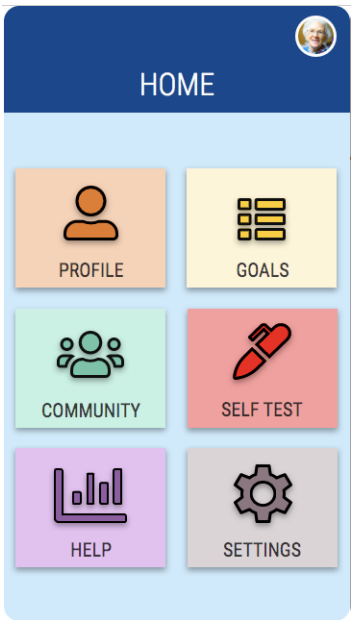


Figure H.6 – Home screen

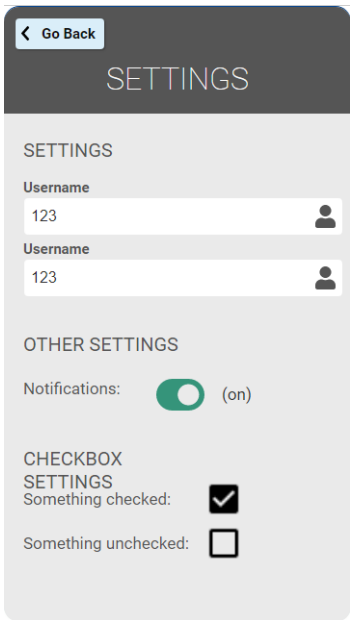


Figure H.7 – Settings screen

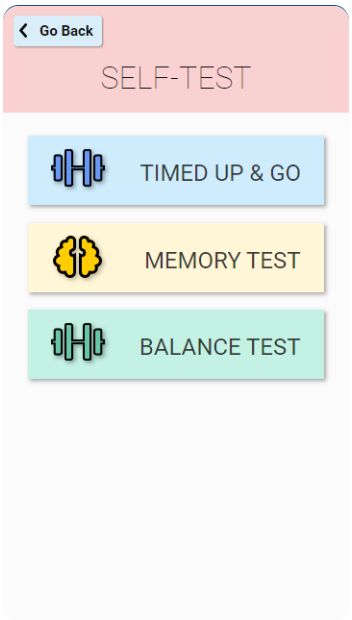


Figure H.8 – Self-test screen

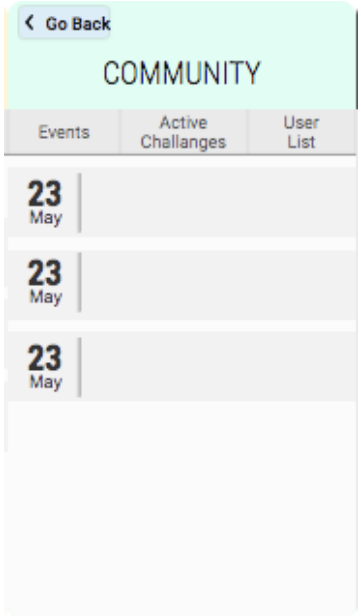


Figure H.9 – Community event screen

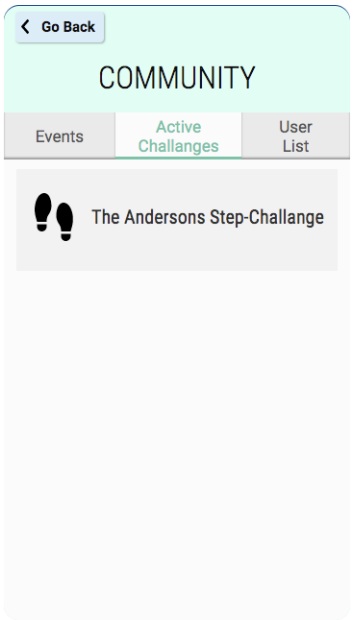


Figure H.10 – Community challenges screen

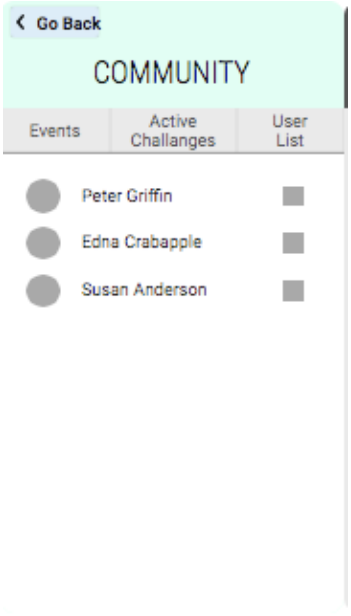


Figure H.11 – User list screen

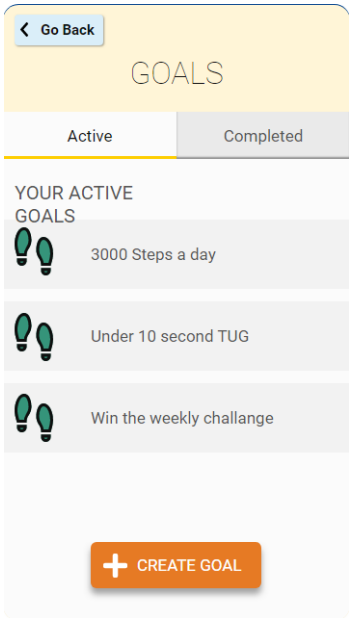


Figure H.12 – Goals: active

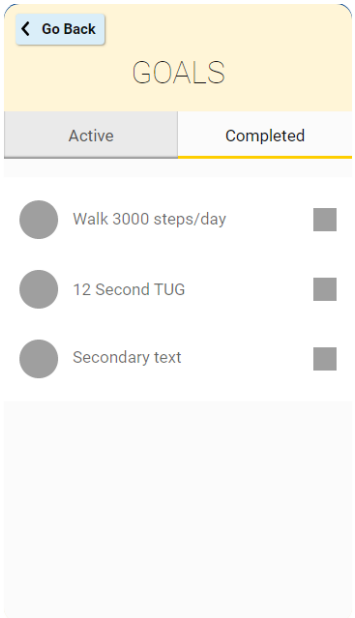


Figure H.13 – Goals: completed

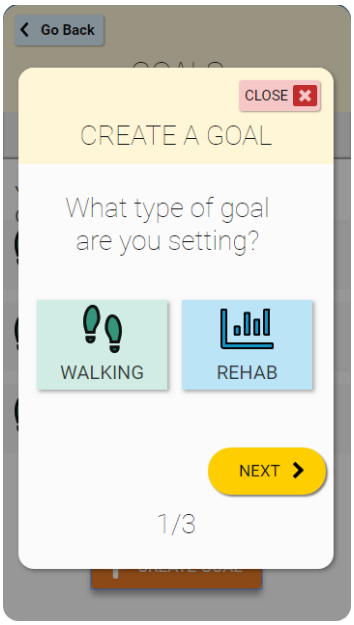


Figure H.14 – Create a goal 1/3

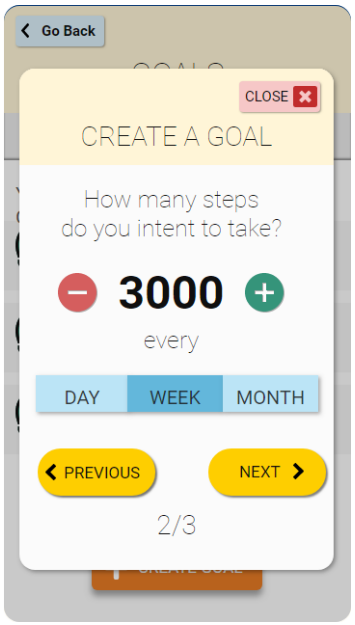


Figure H.15 – Create a goal 2/3

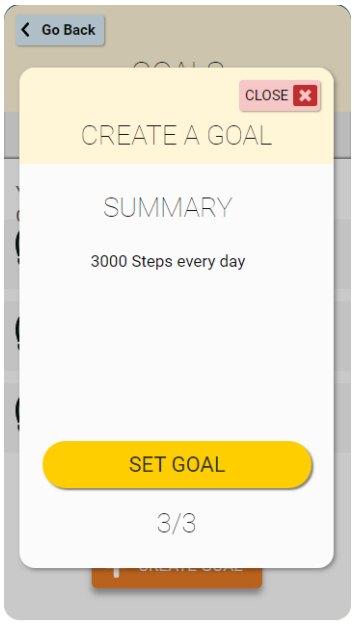


Figure H.16 – Create a goal 3/3

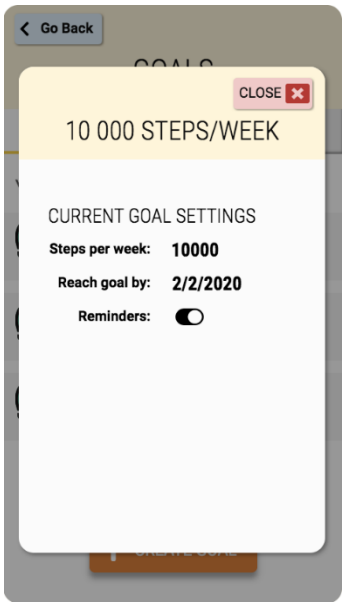


Figure H.17 – Goal settings

Figure H.18 – Goal statistics

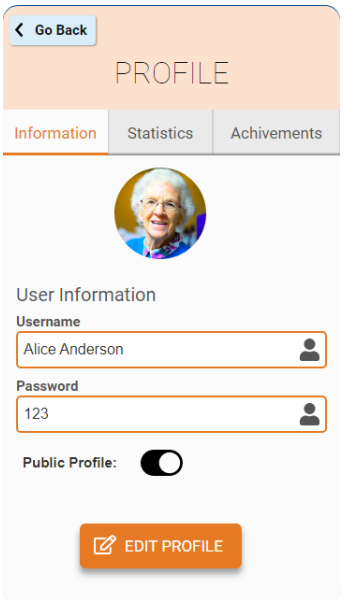


Figure H.19 – Profile: information

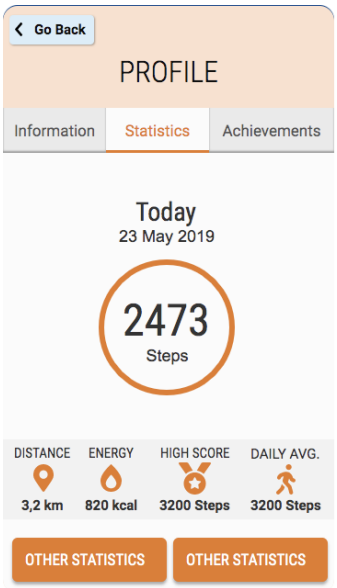


Figure H.20 – Profile: statistics

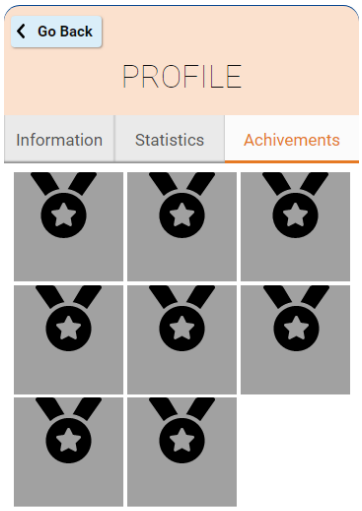


Figure H.21 – Profile: achievements

Revised prototype

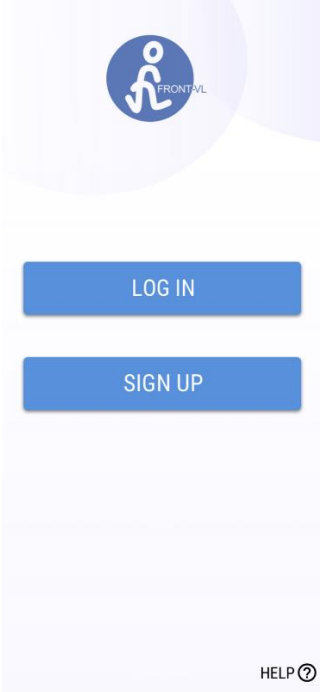


Figure I.1 – Welcome screen

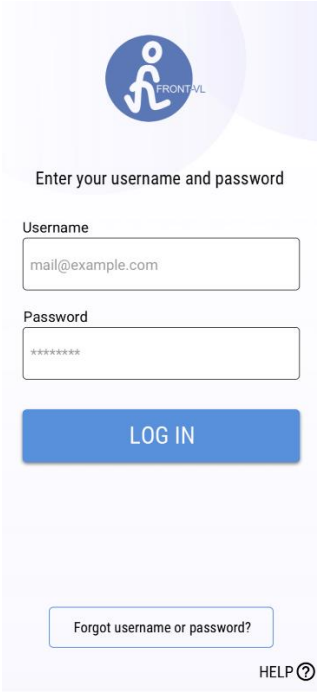


Figure I.2 – Log in screen

Figure I.3 – Home screen

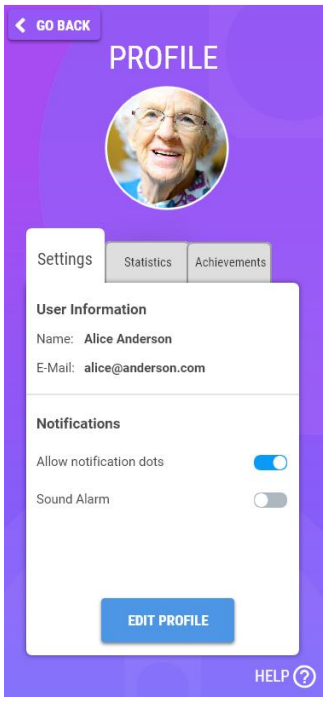


Figure I.4 – Profile: Settings

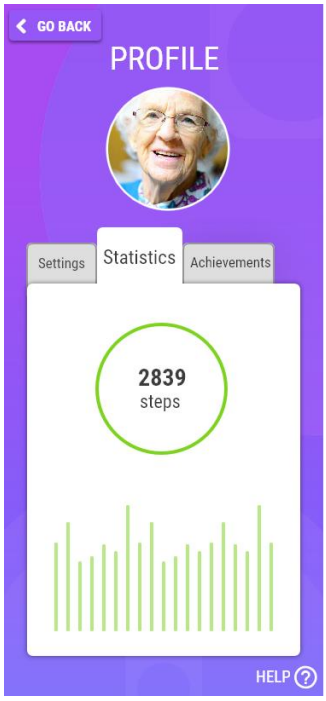


Figure I.5 – Profile: statistics

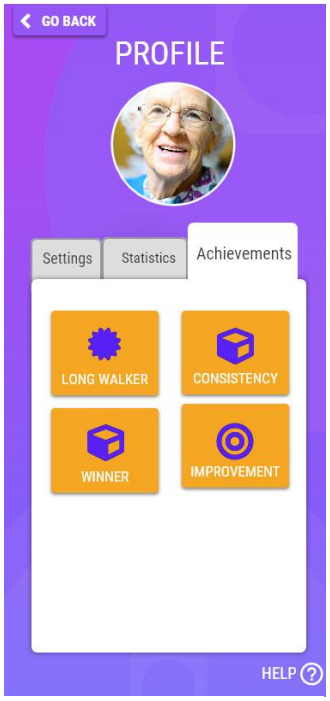


Figure I.6 – Profile: achievements

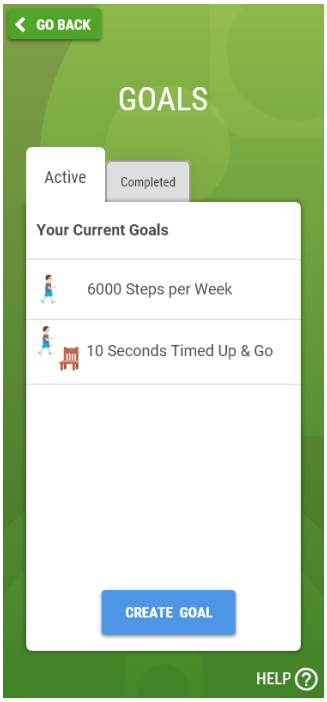


Figure I.7 – Goals: Active

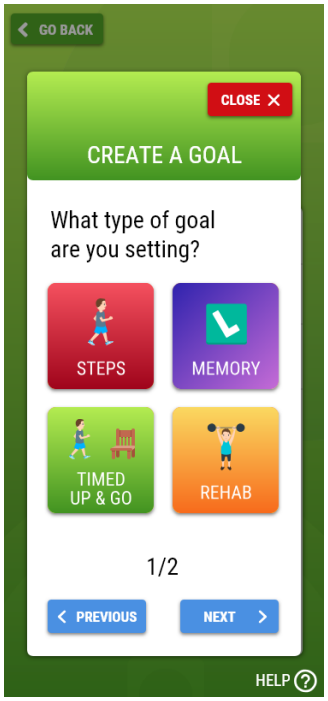


Figure I.8 – Goals: Create a goal (1)

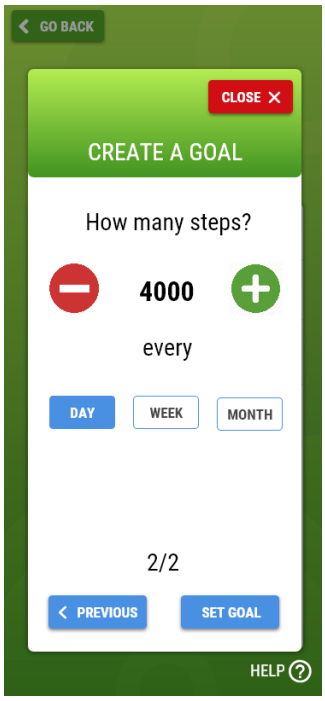


Figure I.9 – Goals: Create a goal (2)

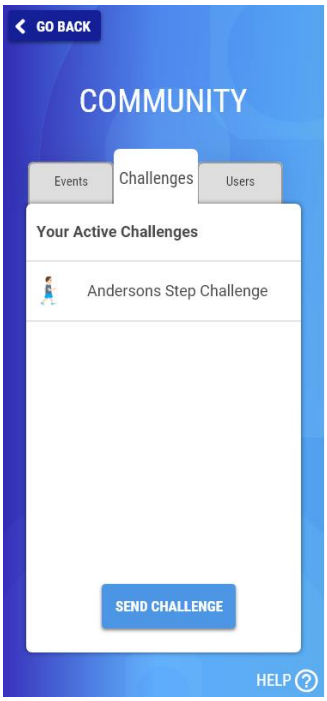


Figure I.10 – Community: Challenges

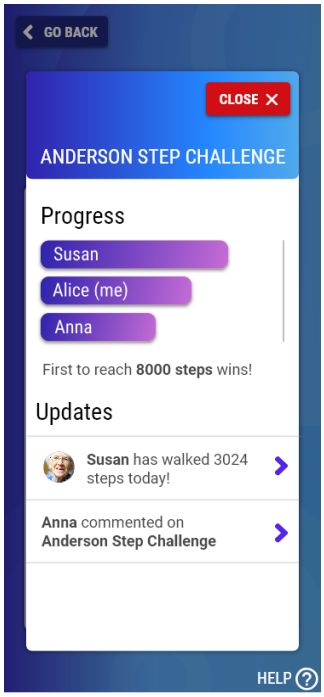


Figure I.11 – Community: Challenges (Challenge details)

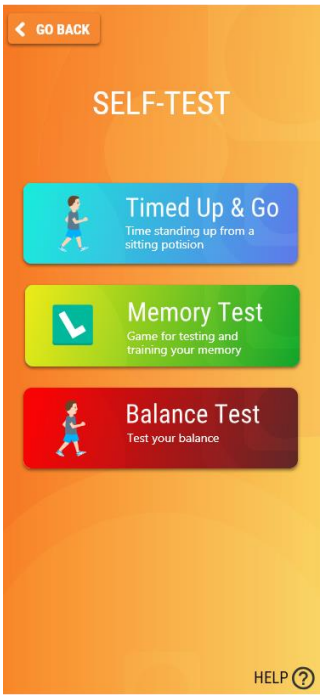


Figure I.12 – Self-test

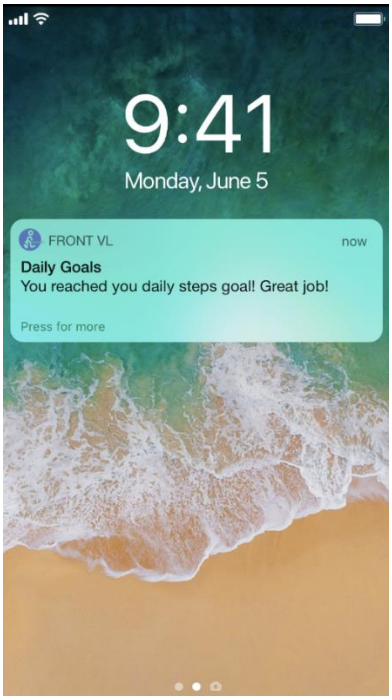


Figure I.13 – Notification: Daily Goals

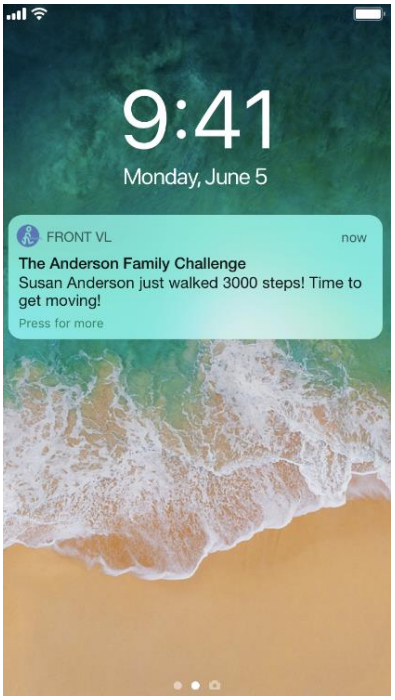


Figure I.14 – Notification: Challenge

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